Information Fusion 139 (2027) 104921



Contents lists available at ScienceDirect

Information Fusion

journal homepage: www.elsevier.com/locate/inffus



Adaptive dual-view score fusion for open-set node classification on heterophilic graphs

Long Xu, Zhiqiang Pan, Honghui Chen *

National Key Laboratory of Information Systems Engineering, Changsha, 410000, China

ARTICLE INFO

Keywords:
Open-set node classification
Graph neural networks
Out-of-distribution detection
Heterophilic graphs
Multi-source score fusion
Adaptive fusion

ABSTRACT

Open-set node classification methods uniformly rely on propagating out-of-distribution signals through graph topology, an assumption that fails on heterophilic graphs where connected nodes carry different labels. We formalize the out-of-distribution signal dilution effect: when cross-distribution edges dominate, neighborhood aggregation attenuates the discriminative gap between in-distribution and out-of-distribution scores. This produces three failure modes that share a single root cause, the blind aggregation of unreliable information sources. To address these failures, we propose HetOOD, a per-node adaptive propagation framework that decouples ego features from neighbor aggregation via learned gating, preventing representation collapse on heterophilic regions. Its dual-view out-of-distribution scoring module combines topology-free and topology-aware signals through a score-consensus weight derived from neighborhood scoring agreement, adapting propagation strength per node without labels. Experiments on 9 benchmarks show consistent improvements over all existing methods, particularly on the most challenging heterophilic datasets.

1. Introduction

Graph neural networks (GNNs) have become the dominant paradigm for learning on graph-structured data, achieving strong performance on node classification [1–3], link prediction [4], and graph-level tasks across diverse domains [5–7]. When graph data contains nodes whose labels fall outside the training set, the task of simultaneously classifying in-distribution (ID) nodes and detecting out-of-distribution (OOD) nodes is termed *open-set node classification* (OSNC) [8–11].

Most OSNC methods adopt a $(C+1)$-class framework that augments the known classes with an "unknown" class, and propagate OOD scores through graph topology via message passing. On homophilic graphs, where connected nodes tend to share labels, energy-based propagation [9,12], score propagation [10], uncertainty propagation [13], and negative mixup augmentation [11] all achieve strong detection. At their core, these methods fuse a node's own OOD evidence with that of its neighbors, and the strategy succeeds precisely because the two sources are positively correlated under homophily.

On heterophilic graphs, where connected nodes frequently carry different labels, the same propagation becomes counterproductive. We identify an *OOD signal dilution effect*. When a node's neighbors belong to different classes, aggregating their OOD scores attenuates rather than enhances discriminability. The lower the homophily ratio h, the stronger the dilution, with near-random OOD detection on Wikipedia-family graphs. Three failure modes follow. On Wikipedia-family graphs, heterophilic message passing collapses most nodes into a single class (representation collapse). On WebKB graphs, the GAT branch underperforms a plain MLP yet OOD scoring relies on it (classifier-OOD disconnect). Across both families, poor initial scores contaminate pseudo-OOD selection in a self-reinforcing loop (pseudo-OOD vicious cycle). What these failures share is a root cause: forcing topology-dependent OOD scoring on graphs where topology is unreliable. Classical fusion theory has long established that naively aggregating unreliable sources degrades performance [14,15], and adaptive strategies that modulate source contributions are the standard remedy. Fig. 1 summarizes these failure modes.

To address these failures, we propose HetOOD (Heterophilic-graph OOD detection), a framework built on per-node adaptive propagation. The first component, Ego-Neighbor Disentangled GAT (ENDGAT), introduces a learned gate $\theta_i \in (0,1)^d$ for each node that decouples ego features from neighbor aggregation: the gate converges toward 0 on homophilic graphs and toward 1 on heterophilic graphs, preventing collapse. The second component, Dual-View OOD Scoring (DVOS), combines a topology-free feature view from an MLP branch with a topology-

* Corresponding author.
E-mail addresses: xulong@nudt.edu.cn (L. Xu), panzhiqiang@nudt.edu.cn (Z. Pan), chenhonghui@nudt.edu.cn (H. Chen).

https://doi.org/10.1016/j.inffus.2026.104921
Received 7 May 2026; Received in revised form 29 August 2026; Accepted 31 October 2026
Available online 3 November 2026
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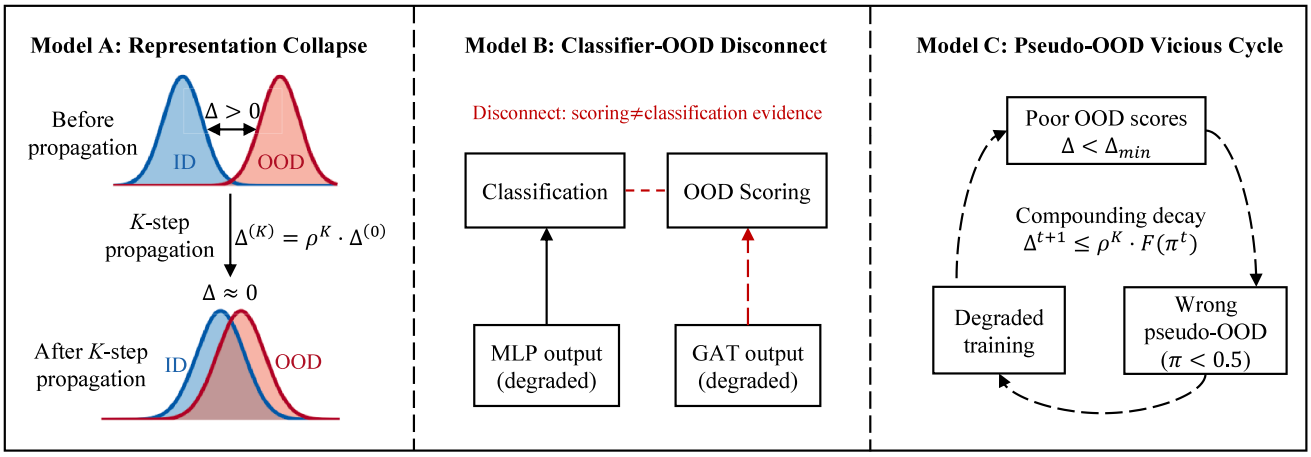


Fig. 1. Failure modes on heterophilic graphs.

aware structure view from ENDGAT, modulated by a Score-Consensus Weight ϕ derived from the variance of MLP OOD scores in each node's neighborhood. With agreeing neighbors, DVOS recovers standard propagation, and disagreement suppresses neighbor influence.

- **Analytical framework.** Three propositions formalize the interaction between graph heterophily and OSNC. Discriminability decays under multi-step propagation on heterophilic graphs, with signal inversion under severe heterophily. Neighborhood score variance detects this decay without labels. The pseudo-OOD feedback loop collapses on heterophilic graphs, and DVOS prevents this collapse through a discriminability invariant.
- **Method.** HetOOD combines ENDGAT, which performs ego-neighbor disentanglement via per-node gating, and DVOS, which performs dual-view OOD scoring with ϕ-modulated neighbor aggregation. The constituent mechanisms have precedents in heterophilic GNNs and in classical fusion theory, but their combination serves a new objective, that of stabilizing OOD scoring when topology is unreliable, and yields, to our knowledge, the first analytical stability guarantee (Proposition 3) for open-set node classification on heterophilic graphs.
- **Evaluation.** Experiments on 9 benchmarks spanning 3 homophilic and 6 heterophilic datasets demonstrate consistent improvements. Heterophilic GNN backbone comparisons isolate the contribution of adaptive scoring from heterophily-aware encoding.

2. Related work

2.1. Open-set node classification and graph OOD detection

Open-set node classification extends traditional node classification by requiring models to handle unseen classes at test time. In real-world graph applications the label space is rarely exhaustive, making this setting a practical necessity rather than a convenience [16].

Early approaches like OpenWGL [17] and OODGAT [8] modeled node uncertainty through variational autoencoders and attention-based ID/OOD separation, respectively. G2Pxy [18] improved OOD detection by generating proxy unknown classes via distribution modeling. A second family of methods propagates OOD scores along graph topology. GNNSAFE++ [9] proposed energy-based belief propagation, establishing theoretical connections between energy functions and OOD detectability. Its successor NODESAFE++ [12] added bounded energy guarantees. GRASP [10] combined edge augmentation with neighborhood shaping for score propagation, while TopoOOD [19] incorporates neighborhood topology and context to reshape node neighborhoods for OOD detection. In a third direction, training is augmented with synthetic OOD samples. negMIX [11], the direct predecessor of this work, introduced negative mixup, which mixes pseudo-OOD embeddings with negated ID embeddings, achieving strong results on homophilic benchmarks. GOLD [20] used adversarial pseudo-OOD generation for the same purpose. More recent methods include EMP [21], which identifies unknown-class nodes through entropy propagation and entropy clustering, DeGEM [22], which decouples feature-based and topology-based energy scoring specifically for heterophilic graph OOD detection, and LECT [23], which uses LLMs to generate pseudo-OOD samples with energy-based contrastive learning for text-attributed graphs.

For node-level OOD detection more broadly, methods can be categorized into post-hoc scoring [24–26], energy-based methods [27,28], and topology-aware propagation [9,19]. Post-hoc scoring applies scoring functions to pre-trained GNN outputs. Energy-based methods formulate detection as energy minimization. Topology-aware methods explicitly propagate OOD signals along edges.

All these methods implicitly assume homophilic graph structure and propagate OOD signals through neighborhoods, which may be counterproductive on heterophilic graphs as we demonstrate. Viewed through the lens of information fusion [14,29], these methods perform *score-level fusion* by combining a node's own OOD score with those of its neighbors. The fusion rule is typically a fixed weighted sum or attention-based aggregation, implicitly assuming that the topology source is always informative. Classical fusion theory has long shown that fixed combination rules degrade when source reliability varies across inputs [15], and adaptive fusion that estimates per-instance source reliability is the standard remedy. Yet no existing OSNC method adapts its fusion strategy based on local source reliability. To our knowledge, no existing work has provided an analytical model characterizing the interaction between graph homophily and OOD detection effectiveness, nor proposed solutions that address the representation collapse and pseudo-OOD vicious cycle that arise on heterophilic graphs.

Beyond OSNC, the broader literature on adaptive information fusion and neighbor selection provides relevant context. Adaptive kNN graph construction [30] reshapes graph topology from local geometry, complementing our node-level adaptation on a fixed topology. Multi-view fusion methods, including quantum-enhanced configurable-kernel multi-view feature selection with adaptive fusion [31] and ambiguity-calibrated multi-view open-set learning with view-wise debiasing [32], study how to adaptively weight complementary views under varying source reliability. The taxonomic review of GNN-based multi-view learning [33] surveys this direction. Robust classification under data heterogeneity and noise is also explored from a quantum-assisted angle, including quantum SVM for noisy data [34] and quantum KNN with k-value and neighbor selection [35], which examine how to select and weight neighbors under irregular sample structure. These works share with HetOOD the principle of source-reliability-aware

aggregation and adaptive neighbor weighting, but operate in non-graph or closed-set settings. HetOOD is the first to instantiate adaptive multi-view fusion for open-set node classification on heterophilic graphs, where the reliability of the topology source varies with local homophily and must be estimated without labels.

2.2. GNNs on heterophilic graphs

H2GCN [36] first identified that standard message passing performs low-pass filtering that is harmful on heterophilic graphs, and proposed separating ego and neighbor representations with concatenation at the final layer. FAGCN [37] introduced edge-level attention that can assign negative weights, enabling per-edge frequency adaptation between low-pass and high-pass filtering. ACM-GCN [38] analyzed spectral properties of heterophilic GNNs and proposed adaptive channel modeling that combines low-pass, high-pass, and identity channels. GPR-GNN [39] learned optimal polynomial coefficients for spectral filtering, allowing the model to adaptively weight different frequency components.

While these advances have significantly improved closed-set classification on heterophilic graphs, none of them address OOD detection. The challenges here are fundamentally different from those in closed-set settings. Message passing and OOD scoring interact in a compounding way: poor representations produce poor OOD scores, which in turn degrade pseudo-OOD selection. Compounding this further, neighbor OOD signal reliability varies dramatically across the graph depending on local homophily, calling for node-level adaptation of neighbor aggregation, which ϕ provides within the structure view, while the coarser-grained balance between the feature and structure views is handled by a global fusion weight. Pseudo-OOD sample construction also becomes more challenging when the model cannot reliably distinguish between ID and OOD nodes in the embedding space.

2.3. Graph contrastive learning

Graph contrastive learning (GCL) learns representations by contrasting different views of the same graph [40–42]. Existing GCL methods construct views through hand-crafted augmentations (node dropping, edge perturbation, diffusion) and focus on representation learning rather than calibration of class probabilities. Applied to OSNC, this creates a mismatch: the augmentations may not preserve discriminative features needed for OOD detection, and the learned representations are not designed to separate ID from OOD nodes. In HetOOD, the two ENDGAT layers naturally produce structurally diverse ego representations without any manual augmentation design, since each layer captures information from different neighborhood orders. We leverage these multi-scale ego projections for cross-layer contrastive learning, which serves as a regularizer to improve intra-class compactness and inter-class separation. Because the contrastive objective operates exclusively on ego projections, the learned discriminative structure is independent of neighbor quality. This connects to the multi-view representation fusion paradigm [43,44], where representations from complementary views are aligned to capture both consistent and complementary structure.

3. Preliminaries

3.1. Notation and problem formulation

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \boldsymbol{X})$ denote a graph, where $\mathcal{V}$ is the node set with $|\mathcal{V}| = N$, $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the edge set, and $\boldsymbol{X} \in \mathbb{R}^{N \times F}$ is the node feature matrix. For node $v_i \in \mathcal{V}$, its feature vector is $\boldsymbol{x}_i \in \mathbb{R}^F$, its label is $y_i \in \mathcal{Y}$, and its neighbor set is $\mathcal{N}_i = \{v_j : (v_j, v_i) \in \mathcal{E}\}$ with degree $d_i = |\mathcal{N}_i|$. We denote the set of labeled nodes as $\mathcal{V}_\ell$ and unlabeled nodes as $\mathcal{V}_u$.

In open-set node classification, the label space is partitioned into ID classes $\mathcal{C}_{\text{ID}} = \{c_1, \ldots, c_C\}$ and OOD classes $\mathcal{C}_{\text{OOD}}$, with $\mathcal{C}_{\text{ID}} \cap \mathcal{C}_{\text{OOD}} = \emptyset$. Only nodes from $\mathcal{C}_{\text{ID}}$ are observed during training. At test time, the model encounters nodes from both $\mathcal{C}_{\text{ID}}$ and $\mathcal{C}_{\text{OOD}}$, and must classify ID nodes into their correct class and detect OOD nodes. We formulate OSNC as a $(C{+}1)$-class classification problem with augmented class set $\mathcal{C}^* = \mathcal{C}_{\text{ID}} \cup \{c_\perp\}$, where $c_\perp$ denotes the unknown class. The model outputs a probability vector $\boldsymbol{q}_i = [q_{i,1}, \ldots, q_{i,C}, q_{i,\perp}] \in \mathbb{R}^{C+1}$ for each node v_i.

3.2. Graph homophily

The edge homophily ratio of a graph is:

$$h(\mathcal{G}) = \frac{|\{(v_i, v_j) \in \mathcal{E} : y_i = y_j\}|}{|\mathcal{E}|} \tag{1}$$

A graph is homophilic when $h \gg 0.5$ and heterophilic when $h \ll 0.5$. The local homophily of node v_i is $h_i = |\{v_j \in \mathcal{N}_i : y_j = y_i\}|/|\mathcal{N}_i|$. We note that alternative homophily measures exist, such as node homophily and adjusted homophily [45]. We use edge homophily for consistency with prior OSNC work [10,11].

4. Motivation analysis

Many OSNC methods compute node-level OOD scores and propagate them through the graph:

$$\omega_i = g(\boldsymbol{e}_i) + \gamma \cdot \text{AGG}\big(\{g(\boldsymbol{e}_j) : v_j \in \mathcal{N}_i\}\big) \tag{2}$$

where $g(\cdot)$ produces a local OOD score, γ controls the propagation strength, and $\text{AGG}(\cdot)$ is a neighborhood aggregation function. Energy-based methods (GNNSAFE++, NODESAFE++) define g via class energies. Score-based methods (GRASP, TopoOOD) propagate learned scores. Uncertainty-based methods (OODGAT, GPN) use predictive entropy. All share the structural assumption that propagating OOD evidence through topology improves detection. On heterophilic graphs, this assumption is counterproductive.

4.1. Analytical model: OOD signal dilution

Definition 1 (OOD Discriminability)**.** For node v_i with local score $g(\boldsymbol{e}_i)$, define the OOD discriminability as the gap between expected scores of OOD and ID nodes:

$$\Delta_g = \mathbb{E}_{v_j \in \text{OOD}}\big[g(\boldsymbol{e}_j)\big] - \mathbb{E}_{v_j \in \text{ID}}\big[g(\boldsymbol{e}_j)\big] \tag{3}$$

Proposition 1 (OOD Signal Dilution)**.** *Let $p_{\text{IO}} = P(y_j \in \mathcal{C}_{\text{ID}} \mid v_j \in \mathcal{N}_i,\ y_i \in \mathcal{C}_{\text{OOD}})$ denote the expected fraction of ID neighbors for an OOD node, and p_{OI} the converse. Under mean aggregation with strength γ for K steps:*

$$\mathbb{E}\left[\Delta_g^{(K)}\right] = \rho^K \cdot \Delta_g^{(0)}, \quad \rho = 1 + \gamma\big(1 - p_{\text{IO}} - p_{\text{OI}}\big) \tag{4}$$

Proof. Under mean aggregation, $\omega_i = g(\boldsymbol{e}_i) + \gamma \cdot |\mathcal{N}_i|^{-1} \sum_{v_j \in \mathcal{N}_i} g(\boldsymbol{e}_j)$. For an OOD node, $\mathbb{E}[\omega_i^{\text{OOD}}] = g^{\text{OOD}} + \gamma[p_{\text{IO}} \cdot g^{\text{ID}} + (1{-}p_{\text{IO}}) \cdot g^{\text{OOD}}]$. Similarly, $\mathbb{E}[\omega_i^{\text{ID}}] = g^{\text{ID}} + \gamma[p_{\text{OI}} \cdot g^{\text{OOD}} + (1{-}p_{\text{OI}}) \cdot g^{\text{ID}}]$. Subtracting gives the one-step discriminability $\Delta_g' = (1 + \gamma(1 - p_{\text{IO}} - p_{\text{OI}})) \cdot \Delta_g$. Since each step applies the same linear transformation, $\Delta_g^{(K)} = \rho^K \cdot \Delta_g^{(0)}$ by induction. □

Let $\eta_{\text{cross}} = p_{\text{IO}} + p_{\text{OI}}$ denote the cross-distribution edge density. When $\eta_{\text{cross}} < 1$ (homophilic), $\rho > 1$ and discriminability grows with propagation. When $1 < \eta_{\text{cross}} < 1 + 1/\gamma$, discriminability decays exponentially. When $\eta_{\text{cross}} > 1 + 1/\gamma$, the signal inverts: OOD nodes appear less anomalous than ID nodes.

Proposition 1 adopts mean aggregation as an analytically tractable model, and the same dilution dynamics apply qualitatively to learned aggregators. Attention-weighted methods such as GAT could in principle reduce cross-distribution influence, but the attention coefficients receive no supervision on same- versus cross-distribution edges and thus cannot resolve this in practice, as we confirm empirically below. The dilution condition $\eta_{\text{cross}} > 1$ characterizes distribution-level mixing, related

to but distinct from class-level homophily h. On strongly heterophilic graphs, high cross-distribution edge density naturally satisfies this condition.

Crucially, the dilution condition $\eta_{\text{cross}} > 1$ can be detected without label information, through neighborhood score variance.

Proposition 2 (Variance as Dilution Detector). *Let $\omega^f \mid y{=}c \sim (\mu_c, \sigma^2)$ with between-class gap $\Delta_\omega > 0$, and let p_i denote the OOD fraction in the neighborhood of v_i. Then:*

$$\mathbb{E}\big[\mathrm{Var}(\mathcal{N}_i)\big] = \sigma^2 + p_i\big(1-p_i\big)\Delta_\omega^2 \tag{5}$$

Furthermore, $\mathbb{E}[\mathrm{Var}(\mathcal{N}_i)]$ is monotonically increasing in p_i for $p_i \in [0, 0.5]$ and monotonically decreasing for $p_i \in [0.5, 1]$.

The proof follows from the law of total variance applied to the two-component mixture in each neighborhood.

Since $p_i(1-p_i)$ is maximized at $p_i = 0.5$ and vanishes at $p_i \in \{0, 1\}$, elevated variance signals neighborhood mixing regardless of which distribution dominates. Because the dilution condition $\eta_{\text{cross}} > 1$ implies mixed neighborhoods, $\mathbb{E}[\mathrm{Var}(\mathcal{N}_i)] > \sigma^2$ is a necessary consequence. Taken together, these two propositions motivate the Score-Consensus Weight ϕ defined below: $\phi_i = \exp(-\mathrm{Var}(\mathcal{N}_i)/\tau_\phi)$ translates the variance signal into a bounded weight in $(0, 1]$, suppressing neighbor propagation when the dilution condition is detected.

Proposition 3 (Pseudo-OOD Feedback Stability). *Under the score model of Proposition 2, let pseudo-OOD selection use a threshold-based rule with precision $\pi^t = P(y_i \in C_{\text{OOD}} \mid v_i \in \mathcal{P}_t^+)$. Under two-component Gaussian scores with discriminability Δ, let $\alpha = \Phi(\Delta/2\sigma)$ where Φ is the standard normal CDF. The precision satisfies:*

$$\pi = \psi(\Delta) = \frac{\alpha \cdot p_{\text{OOD}}}{\alpha \cdot p_{\text{OOD}} + (1-\alpha)\big(1-p_{\text{OOD}}\big)} \tag{6}$$

where p_{OOD} is the true OOD fraction among unlabeled nodes. The function ψ is strictly increasing with $\psi(0) = p_{\text{OOD}} < 0.5$.

Under K-step propagation with per-step decay factor $\rho < 1$ (Proposition 1), the effective discriminability satisfies $\Delta^{t+1} \le \rho^K \cdot F(\pi^t)$, where $F(\cdot) \ge 0$ is non-decreasing in pseudo-OOD precision. Since ψ is strictly increasing with $\psi(0) < 0.5$, there exists $\Delta_{\min} > 0$ such that $\Delta^t < \Delta_{\min}$ implies $\pi^t < 0.5$ (pseudo-OOD precision below random). Beyond this point, decay and degrading pseudo-OOD quality reinforce each other, driving the system toward a low-discriminability fixed point.

Assume that (i) the MLP branch achieves positive discriminability $\Delta_f^* > 0$ from labeled ID features, and (ii) the structure view satisfies $\Delta_s \ge 0$ (OOD nodes receive no lower scores than ID nodes in expectation). Under DVOS fusion (Eq. (18)), the combined discriminability satisfies:

$$\Delta_{\text{DVOS}}^t \ge (1-\beta)\Delta_f^* > 0 \quad \textit{for all sufficiently large } t \tag{7}$$

so $\pi_{\text{DVOS}}^t \ge \psi((1-\beta)\Delta_f^*) \triangleq \pi_{\min} > \psi(0)$, and the compounding decay is prevented.

Proof. Under the two-component Gaussian model, OOD nodes have higher expected scores than ID nodes by Δ. Threshold-based selection achieves precision $\psi(\Delta)$ in Eq. (6), strictly increasing since Φ is strictly increasing. When $\Delta = 0$, $\alpha = 0.5$ and $\pi = p_{\text{OOD}} < 0.5$ (the prior). Setting $\psi(\Delta_{\min}) = 0.5$ yields $\Delta_{\min} > 0$. By Proposition 1, $\Delta^{t+1} \le \rho^K \cdot F(\pi^t)$ where $F \ge 0$ is non-decreasing in precision. When $\Delta^t < \Delta_{\min}$, $\pi^t < 0.5$ and $F(\pi^t)$ provides minimal training benefit, so the system converges to a low-discriminability fixed point.

From DVOS fusion $\omega_i = (1-\beta)\omega_i^f + \beta\omega_i^s$, taking expectations over ID and OOD populations yields $\Delta_{\text{DVOS}} = (1-\beta)\Delta_f + \beta\Delta_s$. By assumption $\Delta_s \ge 0$, so $\Delta_{\text{DVOS}} \ge (1-\beta)\Delta_f$. The MLP branch is trained via $\mathcal{L}_{\text{ce}}$ on labeled ID nodes only, independent of graph structure and $\mathcal{P}^+$, so Δ_f converges to $\Delta_f^* > 0$ determined by feature-space separability. The precision bound $\pi^t \ge \psi((1-\beta)\Delta_f^*) > \psi(0)$ follows from the monotonicity of ψ. □

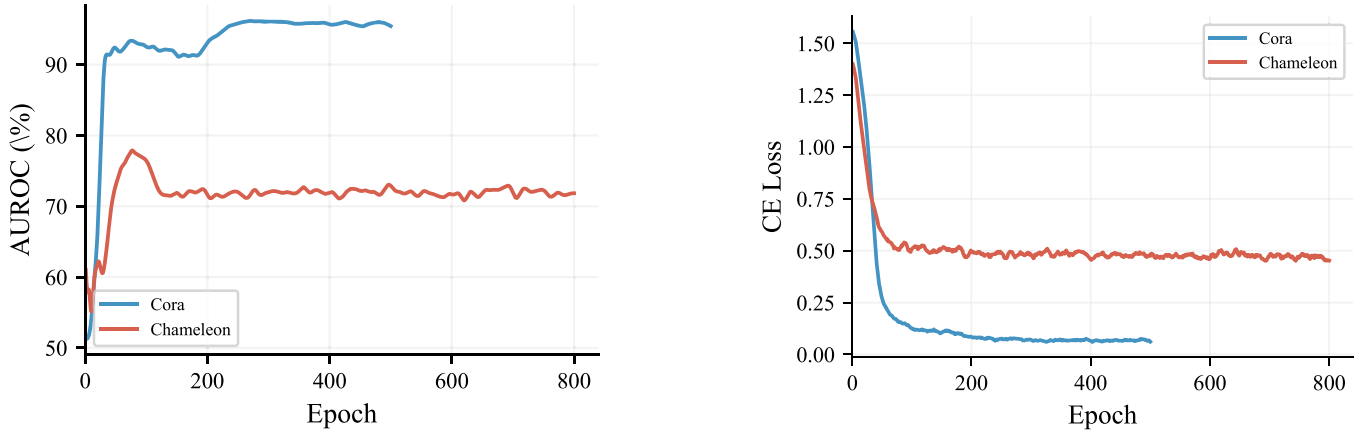
The compounding decay predicted by Proposition 3 explains the training dynamics observed empirically below: OOD AUROC peaks early on Chameleon and then degrades despite continued reduction in classification loss. DVOS breaks this cycle through the invariant $(1-\beta)\Delta_f^*$, which does not depend on pseudo-OOD quality, ensuring pseudo-OOD precision remains above random throughout training. Assumption (i) holds whenever node features carry class information, which is the standard setting in node classification. Assumption (ii) requires the structure view not to actively invert OOD rankings. ENDGAT is designed to prevent representation collapse, and when it functions correctly, $\Delta_s \ge 0$ follows naturally. The invariant thus holds under the condition that ENDGAT succeeds at its design objective, which is strictly weaker than requiring standard message passing to succeed.

On the assumptions. The three propositions rest on assumptions chosen to make the dilution dynamics analytically tractable, and we briefly discuss their applicability and limitations. *(i) Mean aggregation* is used in Proposition 1 as a representative linear aggregator. The decay factor $\rho = 1 + \gamma(1 - p_{\text{IO}} - p_{\text{OI}})$ depends only on the cross-distribution edge density and the propagation strength, not on the specific aggregation weights, so any linear aggregator yields the same qualitative regime of growth under homophily, decay under moderate mixing, and inversion under severe mixing. Learned aggregators such as GAT could in principle attenuate cross-distribution influence, but their attention coefficients receive no supervision distinguishing same- from cross-distribution edges in OSNC, so the dilution condition remains effectively unrecoverable. Our experiments confirm this point, as the GAT-based OSNC methods are among the worst performers on heterophilic graphs in Table 3. *(ii) Two-component Gaussian scores* are used only in Proposition 3 to obtain the closed-form precision $\pi = \psi(\Delta) = \alpha p_{\text{OOD}}/(\alpha p_{\text{OOD}} + (1-\alpha)(1-p_{\text{OOD}}))$ with $\alpha = \Phi(\Delta/2\sigma)$. Proposition 2 requires only finite second moments, since $\mathbb{E}[\mathrm{Var}(\mathcal{N}_i)] = \sigma^2 + p_i(1-p_i)\Delta_\omega^2$ follows from the law of total variance alone and holds for arbitrary score distributions with finite variance. The monotonicity of ψ in Δ, which is the only property used to derive the invariant, does not require Gaussianity either. It holds whenever the score distributions of ID and OOD nodes stochastically dominate each other. *(iii) Positive feature-view discriminability* $\Delta_f^* > 0$ requires that node features carry class information, which is the standard premise of feature-based node classification. We disclose in the Conclusion that this is the fundamental limit of methods without external OOD signals. The propositions therefore provide a *necessary-condition* explanation of heterophilic OSNC failure and a *stability invariant* against the pseudo-OOD vicious cycle, rather than a tight performance bound. Even so, they correctly predict the empirical dynamics of Fig. 2, with an early AUROC peak followed by decay on Chameleon, and the dataset-dependent gains of Table 3, with larger margins on more heterophilic graphs.

4.2. Empirical evidence of failure modes

Fig. 2 shows the training dynamics of negMIX on the homophilic Cora and heterophilic Chameleon datasets, and qualitatively identical patterns were observed for GNNSAFE++ and GRASP on all six heterophilic datasets. Classification loss decreases on both datasets, but OOD AUROC diverges: on Cora it improves steadily, on Chameleon it peaks early and degrades, which is the empirical manifestation of the compounding decay predicted by Proposition 3.

On Wikipedia-family graphs, message passing through heterophilic neighborhoods collapses GAT representations into a single dominant cluster, with over 80% of test nodes classified as one class. Classification loss continues to decrease through collapse while OOD discrimination degrades, rendering standard early stopping counterproductive. On WebKB-family graphs, the GAT branch achieves significantly lower accuracy than a topology-free MLP, yet OOD scoring relies primarily on GAT outputs. Across both families, poor initial OOD scores cause KMeans to misidentify ID nodes as pseudo-OOD and vice versa, creating a self-reinforcing loop. On Chameleon, pseudo-OOD precision drops from 12% to 8% over training, while on Cora it remains above 60%.



(a) OOD AUROC (b) Classification loss

Fig. 2. Training dynamics of negMIX on Cora and Chameleon.

5. Methodology: HetOOD

The preceding analysis motivates three design principles: decouple ego features from neighbor aggregation to prevent collapse, incorporate topology-free OOD scoring alongside topology-aware signals, and make adaptation label-free using only model outputs and structure. HetOOD realizes these through two components, ENDGAT and DVOS, corresponding respectively to the first principle and the remaining two.

5.1. Framework overview

Fig. 3 illustrates the overall architecture of HetOOD. Given a graph $\mathcal{G}$, HetOOD operates through three stages: **Scoring**: a Score-Consensus Weight ϕ_i is computed for each node from the variance of neighborhood MLP OOD scores. **Encoding**: ENDGAT produces ego-neighbor disentangled representations while a parallel MLP branch produces topology-free representations. **OOD Scoring**: DVOS combines topology-free and topology-aware OOD scores through ϕ-modulated neighbor aggregation and a global fusion weight β. The training objective jointly optimizes classification, pseudo-sample augmentation, OOD regularization, and multi-view contrastive learning.

5.2. Score-consensus weight

Designing heterophily-aware methods requires deciding *when* neighborhood aggregation is beneficial for OOD detection. Aggregation is safe for neighborhoods that belong to a single distribution (all ID or all OOD) and harmful for distributionally mixed ones. Rather than estimating this through feature similarity, which is indirect and assumes features carry class information, we propose to measure it directly through the *agreement of topology-free OOD scores* in each node's neighborhood.

Definition 2 (Score-Consensus Weight)**.** For each node v_i, the Score-Consensus Weight is defined as:

$$\phi_i = \exp\left(- \frac{\mathrm{Var}\left(\left\{ \omega_j^f : v_j \in \mathcal{N}_i \right\} \right)}{\tau_\phi} \right) \tag{8}$$

where ω_j^f is the topology-free OOD score from the MLP branch for neighbor v_j, and $\tau_\phi > 0$ is a temperature parameter controlling sensitivity.

Under low neighborhood variance, neighbors agree on their OOD scores, the neighborhood is likely distributionally homogeneous, and aggregation is safe ($\phi_i \approx 1$). Under high variance, neighbors disagree, the neighborhood likely mixes ID and OOD nodes, and aggregation is suppressed ($\phi_i \approx 0$).

Proposition 2 provides the formal basis: under the two-component equal-variance score model assumed there, neighborhood score variance increases with the ID/OOD mixing ratio p_i on $[0, 0.5]$. Since the dilution condition $\eta_{\text{cross}} > 1$ from Proposition 1 implies mixed neighborhoods, elevated variance is a necessary consequence, and ϕ translates this signal into a bounded aggregation weight in $(0, 1]$. Because ϕ is computed from the MLP branch's topology-free scores, it avoids circular dependency with graph aggregation. As training progresses, the between-group gap Δ_ω grows relative to within-group variance σ^2, so ϕ transitions from conservative near-uniform weights to discriminative ones, providing an implicit warm-up without explicit scheduling. The main practical concern is small-degree nodes with $d_i \leq 3$, such as those on WebKB datasets, where the variance estimate is noisy. This is mitigated by the DVOS stability invariant from Proposition 3, which guarantees $(1-\beta)\Delta_f^* > 0$ regardless of ϕ quality, shifting reliance toward the feature view, as confirmed empirically by the degree-wise analysis in Section 6.4.4.

A subtlety of Proposition 2 is that $p_i(1-p_i)$ vanishes at both $p_i = 0$ and $p_i = 1$, so ϕ cannot distinguish a uniformly OOD neighborhood from a uniformly ID one. For an ID node surrounded almost entirely by OOD neighbors, $\phi \approx 1$ would still permit aggregation. This scenario is rare on real heterophilic graphs, which exhibit class mixing rather than complete label inversion, and is mitigated by the feature view's independent counterweight in DVOS fusion.

5.3. ENDGAT: ego-neighbor disentangled GAT

Standard GAT aggregates neighbor information through attention-weighted summation:

$$\boldsymbol{e}_i^{\text{nb}} = \sigma_e\left(\sum_{v_j \in \mathcal{N}_i \cup \{v_i\}} \alpha_{ij} \boldsymbol{W}_{\text{GAT}} \boldsymbol{x}_j \right) \tag{9}$$

where α_{ij} is the attention coefficient and σ_e is the activation. While effective on homophilic graphs, this forces every node to incorporate neighbor information, which degrades representations on heterophilic graphs as demonstrated empirically above.

ENDGAT introduces an adaptive gate that decouples ego features from neighbor aggregation. For each node v_i at layer l, we compute:

$$\boldsymbol{e}_i^{\text{nb}} = \sigma_e\left(\text{GAT}\left(\boldsymbol{e}_i^{(l-1)}, \mathcal{N}_i\right)\right) \tag{10}$$

$$\boldsymbol{e}_i^{\text{ego}} = \sigma_l\left(\boldsymbol{W}_{\text{ego}} \boldsymbol{e}_i^{(l-1)}\right) \tag{11}$$

$$\theta_i = \sigma_s\left(\boldsymbol{W}_{\text{gate}}\left[\boldsymbol{e}_i^{(l-1)} \| \boldsymbol{e}_i^{\text{nb}}\right]\right) \tag{12}$$

$$\boldsymbol{e}_i^{(l)} = \theta_i \odot \boldsymbol{e}_i^{\text{ego}} + \left(1 - \theta_i\right) \odot \boldsymbol{e}_i^{\text{nb}} \tag{13}$$

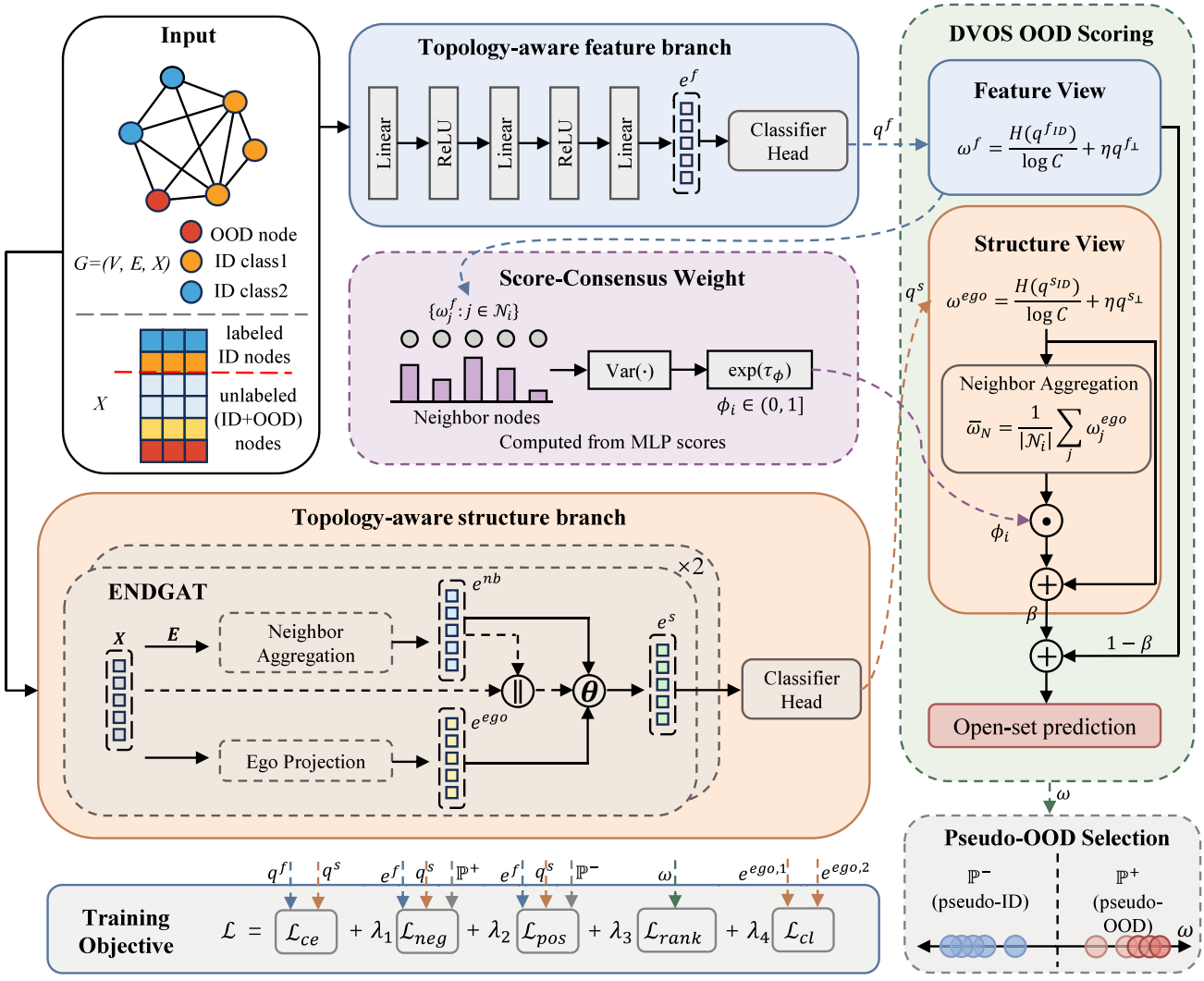


Fig. 3. HetOOD framework. In the topology-aware branch, ‖ denotes concatenation, ⊙ denotes element-wise multiplication, and $\theta_i \in (0,1)^d$ is the learned per-node gate computed from the concatenated representation (Eqs. (12)–(13)).

where e_i^{nb} is the neighbor-aggregated representation, e_i^{ego} is the ego (self) projection, $\theta_i \in (0,1)^d$ is a learnable gate vector, $[\cdot\|\cdot]$ denotes concatenation, $\odot$ is element-wise multiplication, and σ_s is sigmoid.

The gate θ_i is learned end-to-end: on homophilic regions, the gate converges toward 0 (relying on neighbors), while on heterophilic regions, it converges toward 1 (preserving ego features). Two distinctions from prior heterophilic GNN designs are worth noting. H2GCN [36] concatenates ego and neighbor streams at the final layer only, whereas ENDGAT applies node-level gating at every layer, enabling per-layer adaptation to local homophily. FAGCN [37] adapts edge-level attention via frequency-selective weights, whereas ENDGAT's gate operates at the node level based on the joint quality of ego versus neighbor signals, and preserves intermediate ego projections for the contrastive learning objective described below. ACM-GCN [38] likewise adapts at the node level, learning per-node weights over low-pass, high-pass, and identity channels for closed-set classification. ENDGAT's gate instead arbitrates between the ego and neighbor streams themselves rather than among frequency channels, and its ego projections serve double duty as OOD ego scores (Eq. (15)) and as contrastive views. GPR-GNN [39] learns a single set of polynomial filter coefficients shared by all nodes, which is global rather than node-adaptive. In contrast to all four architectures, whose adaptation targets closed-set classification accuracy, ENDGAT's gate is designed jointly for representation quality and OOD-score stability. The condition $\Delta_s \geq 0$ in Proposition 3 relies on it preventing representation collapse. The gate must learn without label-level homophily information, driven solely by the classification and contrastive objectives, which makes its convergence behavior critical to the framework's success.

Two ENDGAT layers are stacked to capture both local and broader neighborhood information. The ego projections e_i^{ego} from *each layer* are retained as separate views for contrastive learning, producing multi-scale ego representations.

The gate learns the correct regime because the classification gradient naturally points toward whichever representation produces lower loss. On heterophilic graphs, ego projections are more discriminative than neighbor-mixed ones, driving $\theta_i \to 1$. The converse holds on homophilic graphs. We verify this empirically in Section 6.4.3.

5.4. DVOS: dual-view OOD scoring with ϕ-adaptive fusion

DVOS realizes the second and third design principles, scoring OOD without topology when topology is unreliable and deciding per node whether neighbor aggregation is safe, at two complementary granularities. The node-level Score-Consensus Weight ϕ_i adaptively modulates neighbor aggregation within the structure view. This is where the dilution effect of Proposition 1 operates, and hence where per-node

adaptation matters most. A global fusion weight β then balances the two views at a coarser granularity, capturing reliability differences between the feature and structure sources that are stable across graphs of similar homophily. A controlled comparison against a per-node learned variant $\beta_i = \sigma(\mathrm{MLP}_\beta(\boldsymbol{x}_i))$ supports this division of labor. The variant matches the global β on Cora but loses 1.2 and 2.9 AUROC points on Chameleon and Texas, where the additional parameters of the gating MLP overfit the small training sets (Section 6.5.1).

5.4.1. Feature view (Topology-free)

An MLP branch processes raw node features $\boldsymbol{x}_i$ independently of graph topology, producing logits $\boldsymbol{l}_i^f \in \mathbb{R}^{C+1}$. The feature-view OOD score is:

$$\omega_i^f = \frac{\mathcal{H}\left(\boldsymbol{q}_i^{f,\mathrm{ID}}\right)}{\log C} + \eta \cdot q_{i,\perp}^f \tag{14}$$

where $\boldsymbol{q}_i^{f,\mathrm{ID}} = \mathrm{softmax}(\boldsymbol{l}_i^{f,1:C})$ is the probability over known classes, $\mathcal{H}(\cdot)$ is the Shannon entropy, $q_{i,\perp}^f$ is the unknown-class probability, and η balances entropy and confidence. The two terms capture complementary OOD signals needed at different training stages. The normalized entropy $\mathcal{H}(\boldsymbol{q}_i^{f,\mathrm{ID}})/\log C$ measures uncertainty among known classes: OOD nodes tend to have high uncertainty. Crucially, this term does not depend on $q_{i,\perp}^f$, which requires learned pseudo-OOD responses. At early epochs when pseudo-OOD quality is low, $q_{i,\perp}^f$ is unreliable, but entropy remains informative because ID nodes are already fitted by $\mathcal{L}_{\mathrm{ce}}$. As training progresses, $q_{i,\perp}^f$ becomes an increasingly strong OOD signal. Together, the two terms ensure discriminative scores throughout training, which is critical for the stability established in Proposition 3. We set $\eta = 1$ throughout all experiments, and performance is stable for $\eta \in [0.5, 2.0]$ since both terms are normalized to $[0, 1]$.

Since the MLP does not use graph topology, ω_i^f is immune to the OOD signal dilution effect (Proposition 1) and provides a reliable baseline score on heterophilic graphs.

5.4.2. Structure view (Topology-aware)

The ENDGAT branch produces logits $\boldsymbol{l}_i^s$ and probabilities $\boldsymbol{q}_i^s$. The structure-view OOD score combines the ego score with *ϕ-weighted* neighbor aggregation:

$$\omega_i^{\mathrm{ego}} = \frac{\mathcal{H}\left(\boldsymbol{q}_i^{s,\mathrm{ID}}\right)}{\log C} + \eta \cdot q_{i,\perp}^s \tag{15}$$

$$\bar{\omega}_{\mathcal{N}_i} = \frac{1}{|\mathcal{N}_i|} \sum_{v_j \in \mathcal{N}_i} \omega_j^{\mathrm{ego}} \tag{16}$$

$$\omega_i^s = \omega_i^{\mathrm{ego}} + \phi_i \cdot \bar{\omega}_{\mathcal{N}_i} \tag{17}$$

The key design choice lies in Eq. (17), where the neighbor aggregation term is scaled by ϕ_i. Under strong neighborhood agreement ($\phi_i \approx 1$), neighbor scores are fully incorporated and standard propagation is recovered. Under disagreement ($\phi_i \approx 0$), neighbor scores are suppressed and dilution is prevented, a departure from methods that uniformly aggregate neighbor OOD signals [9,10]. Unlike multi-step OOD score propagation, the structure view aggregates only neighbors' local ego scores in a single round, deliberately avoiding the discriminability decay characterized by Proposition 1 for $K > 1$.

Because ϕ_i is computed from topology-free MLP scores, it co-evolves with training without trainable parameters or circular dependency with graph aggregation.

On the scale of ω^s. Eq. (17) is additive rather than a convex combination, so the scale of ω_i^s varies with ϕ_i. This is benign because every downstream use of the scores is rank-based. AUROC is invariant to any strictly increasing transformation of the scores, and the pseudo-OOD selection threshold is the 95th percentile of $\{\omega_i\}$ over unlabeled nodes (Section 5.5), which is likewise rank-based. Neither quantity is affected by a positive rescaling of ω^s. The residual scale mismatch in fusion (Eq. (18)) is also bounded: each view score is a sum of at most two terms in $[0, 1]$, so $\omega^f \in [0, 1+\eta]$ and $\omega^s \in [0, 2(1+\eta)]$, a ratio of at most $2\times$ that the validation-selected β absorbs. Finally, the stability invariant of Proposition 3 requires only positive discriminability $\Delta > 0$, a property preserved under any positive rescaling.

5.4.3. ϕ-Adaptive dual-view fusion

The final OOD score combines both views through a global fusion weight $\beta \in [0, 1]$:

$$\omega_i = (1 - \beta) \cdot \omega_i^f + \beta \cdot \omega_i^s \tag{18}$$

Here β controls the overall balance between the topology-free feature view and the topology-aware structure view. Within the structure view, ϕ_i modulates neighbor aggregation, restoring full neighbor propagation under neighborhood agreement and suppressing neighbor influence under disagreement.

5.5. Training objective

The total training objective combines five components:

$$\mathcal{L} = \mathcal{L}_{\mathrm{ce}} + \lambda_1 \mathcal{L}_{\mathrm{neg}} + \lambda_2 \mathcal{L}_{\mathrm{pos}} + \lambda_3 \mathcal{L}_{\mathrm{rank}} + \lambda_4 \mathcal{L}_{\mathrm{cl}} \tag{19}$$

5.5.1. supervised classification $\mathcal{L}_{\mathrm{ce}}$

Cross-entropy loss on labeled ID nodes, applied to both ENDGAT and MLP branches:

$$\mathcal{L}_{\mathrm{ce}} = -\frac{1}{|\mathcal{V}_\ell|} \sum_{v_i \in \mathcal{V}_\ell} \left[\log q_{i,y_i}^s + \frac{1}{2} \log q_{i,y_i}^f \right] \tag{20}$$

The MLP branch receives a reduced weight of $1/2$ because its primary role is to provide topology-free OOD scores for ϕ computation and DVOS fusion rather than to serve as the main classifier. Over-weighting it would encourage the ENDGAT branch to under-utilize graph structure, which is undesirable on homophilic graphs where structure is informative.

5.5.2. Pseudo-sample augmentation

Following [11], we apply 1D KMeans ($k=2$) on DVOS scores $\{\omega_i : v_i \in \mathcal{V}_u\}$ to partition unlabeled nodes, select the higher-scoring cluster filtered at the 95th percentile as pseudo-OOD set $\mathcal{P}^+$, and treat the rest as pseudo-ID set $\mathcal{P}^-$. A key advantage of DVOS in this context is that the topology-free feature view ω_i^f provides more reliable initial scores on heterophilic graphs, breaking the pseudo-OOD vicious cycle.

Negative Mixup $\mathcal{L}_{\mathrm{neg}}$ generates adversarial pseudo-OOD samples by combining pseudo-OOD embeddings with negated ID embeddings. Let $\boldsymbol{r}^+$ denote a sampled pseudo-OOD embedding and $\boldsymbol{r}^-$ a sampled ID embedding. The mixing coefficient κ is drawn from a symmetric Beta distribution:

$$\boldsymbol{r}^* = \kappa \cdot \boldsymbol{r}^+ + (1 - \kappa) \cdot (-\boldsymbol{r}^-), \quad \kappa \sim \mathrm{Beta}\left(\alpha_{\mathrm{neg}}, \alpha_{\mathrm{neg}}\right) \tag{21}$$

The negation $-\boldsymbol{r}^-$ creates a synthetic sample that is maximally distant from ID class y_{r^-} in the embedding space. The soft label for the mixed sample is:

$$\boldsymbol{p}^* = \kappa \cdot \mathbf{1}_\perp + (1 - \kappa) \cdot \left(-\mathbf{1}_{y_{r^-}}\right) \tag{22}$$

where $\mathbf{1}_\perp$ is the one-hot vector for the unknown class and $\mathbf{1}_{y_{r^-}}$ is the one-hot vector for the ID class of r^-. The target $\boldsymbol{p}^*$ in Eq. (22) is a *generalized* soft label whose entries sum to $2\kappa - 1$ rather than one. This deviation from a proper distribution is a design choice rather than an oversight. The positive entry κ pulls the mixed sample toward the unknown class, while the negative entry $-(1-\kappa)$ actively repels it from the generating ID class y_{r^-}. Such an asymmetric push and pull yields sharper OOD boundaries than a standard one-hot unknown-class target, which pulls toward $c_\perp$ without pushing away from y_{r^-}. The cross-entropy in Eq. (23) remains well defined for generalized targets with negative entries because

it is linear in the p_k^*. This design is inherited from negMIX [11] and retained unchanged. The negative mixup loss:

$$\mathcal{L}_{\text{neg}} = -\frac{1}{|\mathcal{P}^+||\mathcal{V}_\ell|} \sum_{\boldsymbol{r}^+ \in \mathcal{P}^+} \sum_{\boldsymbol{r}^- \in \mathcal{V}_\ell} \sum_{k=1}^{C+1} p_k^* \log q_k\left(\Phi(\boldsymbol{r}^*)\right) \tag{23}$$

where $\Phi(\cdot)$ denotes the classification head and $q_k(\cdot)$ the softmax probability for class k. This is applied to both the ENDGAT and MLP branch embeddings.

Positive Mixup $\mathcal{L}_{\text{pos}}$ applies standard mixup between pseudo-ID and labeled ID embeddings to reinforce ID class boundaries. The pseudo-ID embedding $\boldsymbol{r}'^+$ is drawn from $\mathcal{P}^-$ and therefore carries no ground-truth label. We assign it the self-trained label $\hat{y}_{r'^+} = \arg\max_{c \in \mathcal{C}_{\text{ID}}} q^s_{i,c}$ predicted by the ENDGAT branch for the corresponding node v_i. Proposition 3 makes this self-training sound, because pseudo-OOD precision stays above random throughout training. Nodes routed to $\mathcal{P}^-$ are therefore overwhelmingly ID, and their argmax predictions are anchored by the supervised loss $\mathcal{L}_{\text{ce}}$. Let $\boldsymbol{r}'^+$ be a pseudo-ID embedding and $\boldsymbol{r}'^-$ be an ID embedding, with mixed representation $\boldsymbol{r}^\diamond = \kappa \cdot \boldsymbol{r}'^+ + (1-\kappa) \cdot \boldsymbol{r}'^-$ and mixed label $\boldsymbol{p}^\diamond = \kappa \cdot \mathbf{1}_{y_{r'^+}} + (1-\kappa) \cdot \mathbf{1}_{y_{r'^-}}$:

$$\mathcal{L}_{\text{pos}} = -\frac{1}{|\mathcal{P}^-||\mathcal{V}_\ell|} \sum_{\boldsymbol{r}'^+ \in \mathcal{P}^-} \sum_{\boldsymbol{r}'^- \in \mathcal{V}_\ell} \sum_{k=1}^{C} p_k^\diamond \log q_k\left(\Phi(\boldsymbol{r}^\diamond)\right) \tag{24}$$

Stability of the pseudo-OOD cycle. The pseudo-OOD selection creates a feedback loop that can become unstable on heterophilic graphs. Proposition 3 guarantees that DVOS prevents this collapse through the invariant $\Delta_{\text{DVOS}} \geq (1-\beta)\Delta_f^* > 0$, which depends only on labeled ID data and ensures pseudo-OOD precision remains bounded above random throughout training.

5.5.3. OOD score regularization $\mathcal{L}_{\text{rank}}$

We encourage OOD scores of pseudo-OOD nodes to be higher than those of ID nodes through a mean-difference regularization:

$$\mathcal{L}_{\text{rank}} = \frac{1}{|\mathcal{V}_\ell|} \sum_{v_i \in \mathcal{V}_\ell} \omega_i - \frac{1}{|\mathcal{P}^+|} \sum_{v_j \in \mathcal{P}^+} \omega_j \tag{25}$$

This directly optimizes the OOD score discriminability Δ_g defined earlier, encouraging a large gap between ID and pseudo-OOD scores. We use the unclipped mean difference rather than a hinge loss to avoid tuning a fixed margin against the evolving score scale. Since ω_i incorporates DVOS with ϕ-modulated neighbor aggregation, this regularization is more effective on heterophilic graphs than applying it to purely propagation-based scores.

5.5.4. Multi-view contrastive learning $\mathcal{L}_{\text{cl}}$

Contrastive learning is performed on the ego projections $\boldsymbol{e}_i^{\text{ego},(l)}$ from different ENDGAT layers, which form structurally diverse views since each layer aggregates from a different neighborhood order. Class prototypes are computed for each layer:

$$\boldsymbol{p}_k^{(l)} = \frac{1}{|\mathcal{V}_\ell^k|} \sum_{v_i \in \mathcal{V}_\ell^k} \boldsymbol{e}_i^{\text{ego},(l)} \tag{26}$$

and both prototype-to-prototype (P2P) and node-to-prototype (N2P) contrastive losses are applied with temperature τ:

$$\mathcal{L}_{\text{cl}} = \mathcal{L}_{\text{P2P}} + \mathcal{L}_{\text{N2P}} \tag{27}$$

The P2P loss aligns class prototypes across layers:

$$\mathcal{L}_{\text{P2P}} = -\sum_{k=1}^{C} \log \frac{\exp\left(\text{sim}\left(\boldsymbol{p}_k^{(1)}, \boldsymbol{p}_k^{(2)}\right)/\tau\right)}{\sum_{k'=1}^{C} \exp\left(\text{sim}\left(\boldsymbol{p}_k^{(1)}, \boldsymbol{p}_{k'}^{(2)}\right)/\tau\right)} \tag{28}$$

where $\text{sim}(\boldsymbol{a}, \boldsymbol{b}) = \boldsymbol{a}^\top \boldsymbol{b}/(\|\boldsymbol{a}\|\|\boldsymbol{b}\|)$ denotes cosine similarity. The N2P loss pulls node representations toward their class prototypes:

$$\mathcal{L}_{\text{N2P}} = -\frac{1}{|\mathcal{V}_\ell|} \sum_{v_i \in \mathcal{V}_\ell} \sum_{l=1}^{2} \log \frac{\exp\left(\text{sim}\left(\boldsymbol{e}_i^{\text{ego},(l)}, \boldsymbol{p}_{y_i}^{(\bar{l})}\right)/\tau\right)}{\sum_{k=1}^{C} \exp\left(\text{sim}\left(\boldsymbol{e}_i^{\text{ego},(l)}, \boldsymbol{p}_k^{(\bar{l})}\right)/\tau\right)} \tag{29}$$

where $\bar{l} = 3 - l$ denotes the opposite layer. Operating exclusively on ego projections ensures that the discriminative signal is independent of neighbor quality. Algorithm 1 summarizes the overall training procedure.

Algorithm 1 HetOOD training.

Require: Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \boldsymbol{X})$, labels $\boldsymbol{Y}_\ell$, hyperparameters $\lambda_1, \ldots, \lambda_4$

for epoch $t = 1, \ldots, T$ **do**
  Compute MLP outputs: representations $\boldsymbol{e}_i^f$, logits $\boldsymbol{l}_i^f$
  Compute Score-Consensus Weight ϕ_i for all $v_i \in \mathcal{V}$ via Eq. (8)
  Compute ENDGAT outputs: ego views $\{\boldsymbol{e}_i^{\text{ego},(l)}\}$, fused representations $\boldsymbol{e}_i$, logits $\boldsymbol{l}_i^s$
  Compute DVOS scores: ω_i via Eqs. (14)–(18)
  Select $\mathcal{P}^+, \mathcal{P}^-$ via clustering-then-ranking on $\{\omega_i : v_i \in \mathcal{V}_u\}$
  Construct negative and positive mixup samples
  Compute $\mathcal{L}$ via Eq. (19) and update by Adam
end for

Ensure: Trained model, final predictions via $\arg\max_{c \in \mathcal{C}^*} q_{i,c}$

5.6. Complexity analysis

Let N be the number of nodes, E the number of edges, F the feature dimension, d the hidden dimension, and K the number of attention heads.

ENDGAT. Each ENDGAT layer incurs three costs beyond standard GAT: the ego projection $\boldsymbol{W}_{\text{ego}}$ costs $\mathcal{O}(NFd)$, the gate computation $\boldsymbol{W}_{\text{gate}}$ costs $\mathcal{O}(N(F+d)d)$, and the element-wise gating itself is $\mathcal{O}(Nd)$. The GAT attention computation remains $\mathcal{O}(EdK)$. The total for two layers is $\mathcal{O}(EdK + NFd)$, which has the same asymptotic order as standard GAT but with a constant factor of approximately 2× due to the additional ego and gate projections. In practice, this overhead is small relative to the GAT attention computation on sparse graphs where $E \gg N$.

MLP branch. Adds $\mathcal{O}(NFd + Nd^2)$, linear in N.

DVOS. Entropy computation costs $\mathcal{O}(NC)$ and neighbor aggregation costs $\mathcal{O}(E)$. The Score-Consensus Weight computation costs $\mathcal{O}(E)$ per epoch (computing neighborhood variance of MLP scores), and β-weighted fusion adds only $\mathcal{O}(N)$.

Contrastive learning. Prototype computation costs $\mathcal{O}(Nd)$ per view. P2P and N2P losses cost $\mathcal{O}(C^2\, d + NCd)$.

The per-epoch complexity is $\mathcal{O}(EdK + NFd + NCd)$, which is linear in graph size. Compared to standard GAT, HetOOD adds the MLP branch and the ego/gate projections, resulting in a constant-factor overhead of 1.3–1.7× in practice depending on graph density, as detailed in Table 6.

6. Experiments

6.1. Experimental setup

Datasets. We evaluate on 9 benchmark node classification datasets spanning the full homophily spectrum, summarized in Table 1. Three are citation networks with strong homophily: Cora, Citeseer with $h > 0.7$, and PubMed with h=0.80. Six are heterophilic: three Wikipedia-family graphs, Actor, Chameleon, and Squirrel with $h \approx 0.22$, where nodes are web pages connected by hyperlinks, and three WebKB graphs, Texas, Cornell, and Wisconsin with $h < 0.25$, where nodes are university web pages.

Baselines. We compare HetOOD with ten representative OSNC methods. Energy-based propagation methods include GNNSAFE++ [9] and NODESAFE++ [12]. Score propagation methods include GRASP [10] and EMP [21]. Data augmentation methods include negMIX [11] and GOLD [20]. General OSNC frameworks include OpenWGL [17], OODGAT [8], and G2Pxy [18]. We also include DeGEM [22], which decouples feature-based and

Table 1
Dataset statistics.

Dataset	$\lvert\mathcal{V}\rvert$	$\lvert\mathcal{E}\rvert$	$\lvert C\rvert$	F	h
Cora	2708	10,556	7	1433	0.81
Citeseer	3327	9104	6	3703	0.74
PubMed	19,717	88,648	3	500	0.80
Actor	7600	33,391	5	932	0.22
Chameleon	2277	36,101	5	2325	0.23
Squirrel	5201	217,073	5	2089	0.22
Texas	183	325	5	1703	0.11
Cornell	183	298	5	1703	0.13
Wisconsin	251	515	5	1703	0.21

topology-based energy scoring for heterophilic graph OOD detection, as the most directly comparable baseline.

Evaluation Metrics. We evaluate with two groups of metrics. For classification, we report Accuracy and Macro-F1 over all C+1 classes. For OOD detection, we report AUROC and FPR@95, treating OOD detection as binary classification between ID and OOD nodes.

Evaluation Protocol. Following the standard protocol [10,11], we designate a fixed subset of classes as ID (4 classes for Cora, 3 for Citeseer/Actor/Chameleon/Squirrel, 2 for PubMed, and 2 for Texas/Cornell/Wisconsin), treating the remaining classes as OOD. We use 10% of ID nodes for training (20% for the Wikipedia-family graphs and 30% for WebKB datasets, due to their smaller per-class sizes), 10% for validation, and the rest for testing. All results are reported as mean ± standard deviation over 5 random seeds.

Implementation Details. HetOOD uses 2-layer ENDGAT with hidden dimension d=16 and K=2 attention heads (K=4 on Cora and Citeseer). The MLP branch uses a 2-layer architecture with the same hidden dimension 16, ReLU activation, and dropout rate 0.5. Both branches use ELU as the intermediate activation σ_e and σ_l. We use the Adam optimizer with learning rate 0.01 and weight decay 10^{-3}, training for 200 epochs with early stopping on validation accuracy (patience 20). The total loss is balanced by λ_1=1.0, λ_2=0.5, λ_3=0.1, λ_4=0.1. The negative mixup Beta parameter is α_{neg}=0.5 (symmetric Beta), and we sample one negative mixup pair per labeled node per epoch. The contrastive temperature is τ=1.0 on all datasets except squirrel (τ=0.5). The Score-Consensus Weight temperature $\tau_\phi = 0.1$ is chosen so that typical MLP score variances (Var $\sim$ 0.01–0.1) map to a meaningful ϕ range (0.90–0.37): too large a τ_ϕ would flatten all ϕ to 1 (no adaptation), while too small would overfragment the signal. The OOD score balancing parameter is set to $\eta = 1$. The DVOS fusion weight β is selected from $\{0.4, 0.6, 0.8\}$ via the overall C+1-class classification accuracy on the validation set. We use this coarse grid to reduce validation-set overfitting risk on small datasets, and the sensitivity analysis confirms that performance varies by less than 2 AUROC points for every $\beta \in [0.2, 0.8]$ (Section 6.5.1). The pseudo-OOD selection threshold is set to the 95th percentile of OOD scores on unlabeled nodes. For all baselines, we use the official implementations with recommended hyperparameters, retraining on our evaluation protocol for fair comparison.

6.2. Main results

Table 2 reports classification results (Accuracy and Macro-F1), and Table 3 reports OOD detection results (AUROC and FPR@95).

HetOOD achieves the highest accuracy and AUROC on all nine datasets. On homophilic graphs, accuracy improvements over the strongest baseline are moderate, confirming that ego-neighbor disentanglement introduces no penalty when standard aggregation suffices. On heterophilic graphs, the gains are substantially larger, with the widest margins on the WebKB family. DeGEM, the only existing method specifically designed for heterophilic graph OOD detection, achieves competitive classification on Wikipedia-family datasets but underperforms HetOOD across all six heterophilic benchmarks. For OOD detection, HetOOD outperforms the second-best method by a consistently wider margin on heterophilic graphs than on homophilic graphs. Notably, DeGEM achieves AUROC values close to HetOOD on several heterophilic datasets, yet its FPR@95 is substantially higher across all six, indicating poorly calibrated detection scores despite reasonable ranking performance.

The AUROC gains on heterophilic graphs are especially meaningful because most existing methods operate near random on the most challenging datasets. Fig. 4 illustrates why. On homophilic Cora, both HetOOD and negMIX produce well-separated ID/OOD score distributions. On heterophilic Chameleon, negMIX's distributions overlap heavily because propagation through heterophilic edges erodes the score gap, while HetOOD's DVOS fusion maintains clear separation because the topology-free feature view is unaffected by graph structure. Consistent with the dilution analysis of Proposition 1, the improvement comes from avoiding score contamination, not from better representations alone.

6.3. Comparison with heterophilic GNN backbones

Does the improvement of HetOOD stem from its heterophily-aware scoring mechanism (DVOS) or simply from using a heterophily-compatible encoder? To isolate the two, the standard GAT encoder in two representative OSNC methods, GNNSAFE++ and negMIX, is replaced with established heterophilic GNN architectures: H2GCN [36] and FAGCN [37]. All other components (OOD scoring, training objectives, pseudo-sample generation) are kept identical to the original methods.

Replacing GAT with heterophilic GNN backbones improves OSNC performance on most datasets, confirming that heterophily-aware encoding is beneficial. An interesting exception is Texas and Cornell, where FAGCN slightly underperforms H2GCN, likely because its frequency-adaptive parameters overfit on the smallest graphs. Across all six datasets, even the best backbone variant falls short of HetOOD, demonstrating that heterophily-aware encoding alone is necessary but insufficient. The gap is largest on Squirrel, where extreme graph density causes severe OOD signal dilution that backbone replacement cannot address.

Does HetOOD itself depend on ENDGAT? Table 4 therefore also includes two variants in which ENDGAT is replaced by ACM-GCN and by GPR-GNN, the two strongest adaptive architectures discussed in Section 5.3. Both variants remain above every baseline variant on all six datasets, and the differences relative to the ENDGAT version stay within 1.1 AUROC, so the detection load is carried by the dual-view scoring rather than by the encoder. The wins also land where Section 5.3 predicts. GPR-GNN, whose learnable spectral filter is known to excel on the Squirrel and Chameleon pair, edges ahead there, at 69.94 versus 69.47 and 80.41 versus 80.17, both within seed noise. ACM-GCN, whose node-wise channel mixing is the closest analog of ENDGAT's gate, does so on Wisconsin at 84.60 versus 84.36, the smallest WebKB graph and the setting where channel mixing is most at home. ENDGAT keeps the overall edge where node-local gating matters most, on the dense co-occurrence graph Actor at 66.68 versus 65.73, as well as on Texas and Cornell. Averaged over the six datasets, the ENDGAT version remains ahead of both, by +0.60 over ACM-GCN and +0.32 over GPR-GNN, and stays the default. The encoder slot is nonetheless clearly interchangeable among state-of-the-art heterophilic architectures, which is itself evidence that the contribution lies in the scoring framework rather than the encoder.

Table 4 reports AUROC on six heterophilic datasets.

6.4. Component analysis

6.4.1. Ablation study

Nine ablation variants isolate the contribution of every component of HetOOD. "w/o ENDGAT" replaces ENDGAT with standard GAT (no ego-neighbor disentanglement gate). "w/o DVOS" removes the MLP branch entirely, scoring OOD from the structure view alone (ENDGAT retained). "w/o Structure View" scores OOD from the feature view alone ($\omega_i = \omega_i^f$),

Table 2
Classification results. Best and second-best in **bold** and underline. Mean ± std over 5 seeds.

Accuracy ↑ (%) Method	Cora	Citeseer	PubMed	Actor	Chameleon	Squirrel	Texas	Cornell	Wisconsin
OpenWGL	69.07±1.70	68.37±0.98	74.96±0.77	25.78±2.51	33.79±2.53	17.31±2.29	58.24±3.02	43.53±3.19	61.21±2.82
OODGAT	71.69±4.63	67.70±4.30	71.58±3.45	29.28±3.79	31.53±3.16	19.82±2.83	66.34±3.47	48.18±3.52	67.53±3.22
G2Pxy	70.37±2.70	69.43±6.39	62.31±1.62	24.09±3.52	30.23±3.13	15.62±2.59	50.83±3.19	41.79±3.46	55.29±2.96
GNNSAFE++	77.68±2.81	71.58±2.33	77.72±2.16	30.81±2.52	40.83±2.27	20.12±2.09	67.49±2.84	55.32±2.98	70.79±2.52
EMP	75.45±1.69	70.66±0.96	73.66±4.14	28.48±1.91	38.53±1.92	18.22±1.54	60.48±2.18	50.62±2.29	68.17±1.97
GRASP	78.89±2.20	71.36±1.88	78.44±3.99	31.46±2.38	43.54±1.99	21.82±1.79	69.77±2.49	48.51±2.79	72.34±2.31
NODESAFE++	79.74±2.04	72.90±1.59	78.76±2.32	32.83±2.13	41.19±1.77	22.51±1.51	71.21±2.28	63.51±2.47	73.54±1.99
GOLD	76.74±2.23	69.45±2.90	74.11±2.43	28.78±2.69	37.82±2.12	17.76±1.73	63.77±2.37	52.29±2.56	66.78±2.19
DeGEM	80.57±1.73	76.80±1.14	82.62±1.21	45.99±2.18	55.10±2.87	43.40±2.31	65.89±4.57	59.70±3.82	69.57±5.73
negMIX	84.29±0.82	76.47±0.95	84.21±0.71	32.28±1.49	45.34±1.18	23.48±1.27	74.48±1.97	62.76±2.17	75.78±1.82
HetOOD (Ours)	**86.16**±0.46	**83.03**±1.47	**85.32**±1.21	**48.47**±0.96	**57.23**±0.91	**46.17**±0.52	**79.97**±1.76	**69.16**±1.97	**79.62**±1.49
Macro-F1 ↑ (%) Method	Cora	Citeseer	PubMed	Actor	Chameleon	Squirrel	Texas	Cornell	Wisconsin
OpenWGL	67.99±2.12	47.44±2.39	73.60±0.81	13.19±3.84	21.18±3.54	8.82±2.11	42.34±3.64	35.19±3.63	60.84±3.03
OODGAT	75.20±2.48	54.53±2.09	75.73±1.24	15.49±2.78	19.81±3.13	10.49±2.38	48.82±3.84	39.76±3.77	67.22±3.26
G2Pxy	66.21±4.00	50.76±2.54	61.49±1.57	12.33±4.16	18.49±3.84	7.94±2.26	37.53±3.92	33.46±3.92	55.11±3.18
GNNSAFE++	77.04±2.50	55.79±4.50	77.49±2.14	16.43±3.17	26.18±2.58	10.16±1.92	50.81±3.51	46.84±3.48	70.32±3.04
EMP	71.89±2.57	47.75±2.85	73.30±4.85	14.76±2.57	24.27±2.82	9.23±1.68	44.17±3.26	42.33±3.16	67.81±2.49
GRASP	77.17±2.34	55.25±2.27	77.30±4.49	16.88±2.99	28.37±2.33	11.27±1.61	53.23±3.18	40.49±3.26	71.77±2.79
NODESAFE++	78.81±2.22	57.70±1.86	78.18±2.09	17.63±2.84	26.78±2.09	11.81±1.41	54.52±3.04	52.77±3.13	73.13±2.61
GOLD	76.37±1.93	55.22±2.41	73.68±2.25	14.49±3.39	23.52±2.39	9.03±1.54	46.51±3.11	43.79±3.04	66.46±2.71
DeGEM	77.64±2.47	59.77±1.86	82.49±0.97	30.18±2.78	44.14±4.25	31.92±3.42	52.76±5.19	51.84±4.87	66.06±4.21
negMIX	82.78±0.87	59.25±1.67	83.35±0.70	18.03±2.08	29.62±1.51	11.98±1.17	56.29±2.84	53.81±2.81	75.29±2.33
HetOOD (Ours)	**85.61**±0.79	**77.33**±1.31	**83.83**±1.48	**33.76**±2.52	**46.27**±1.22	**34.13**±0.53	**61.06**±1.96	**59.02**±2.23	**79.57**±1.97

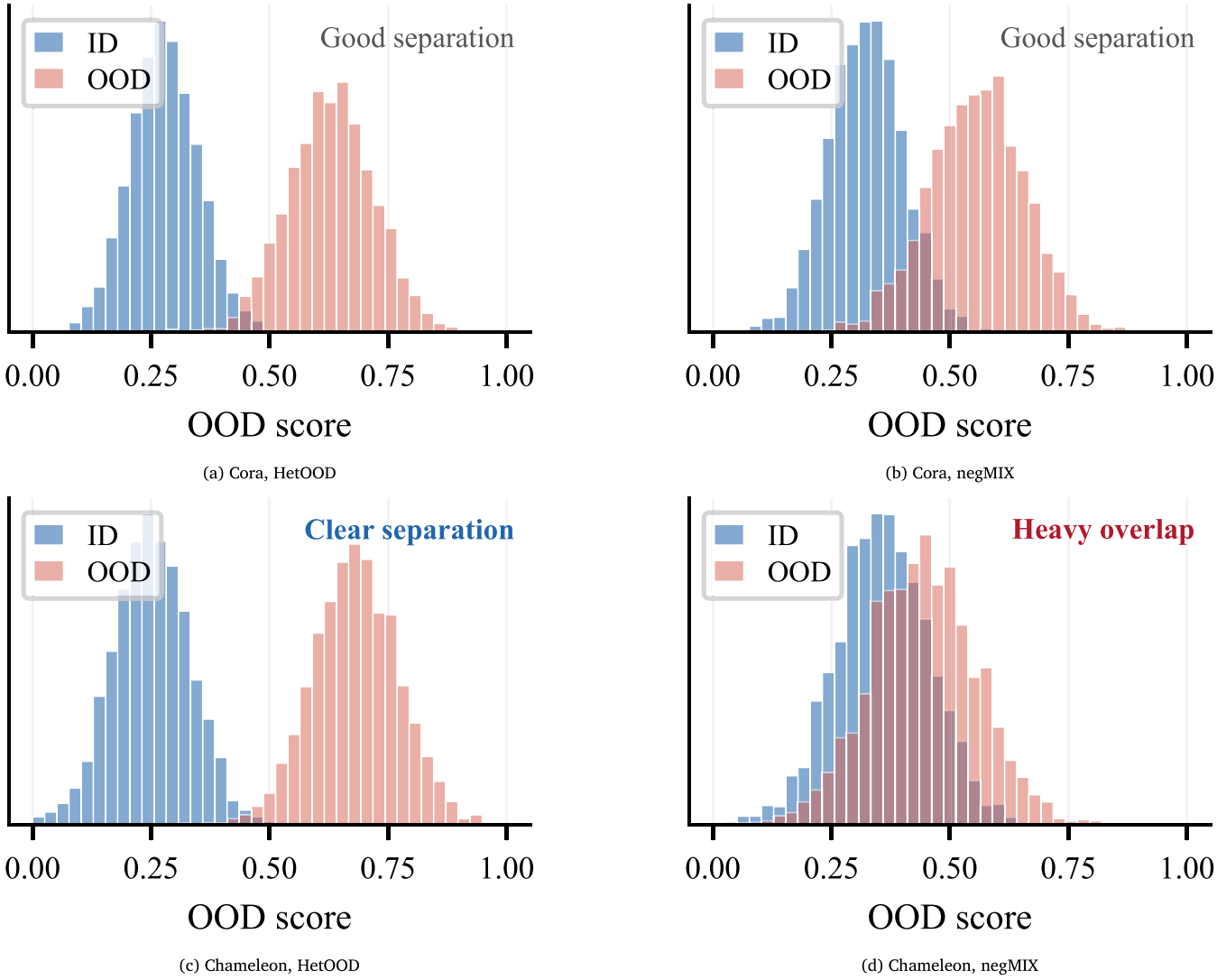
Table 3
OOD detection results. Best and second-best in **bold** and underline. Mean ± std over 5 seeds.

AUROC ↑ (%) Method	Cora	Citeseer	PubMed	Actor	Chameleon	Squirrel	Texas	Cornell	Wisconsin
OpenWGL	81.21±1.75	75.09±2.12	81.46±0.90	52.33±2.72	60.79±2.47	43.51±1.96	60.21±2.89	61.83±2.88	66.49±2.54
OODGAT	88.94±1.79	80.72±3.22	82.45±1.91	56.77±3.04	63.17±2.77	48.61±2.18	68.53±3.17	65.29±3.19	72.78±2.81
G2Pxy	85.40±1.74	80.37±1.68	78.40±1.72	48.47±2.88	58.52±2.73	40.22±2.32	56.29±3.06	57.51±3.13	62.28±2.71
GNNSAFE++	90.38±3.23	83.90±2.00	86.62±2.02	59.19±2.78	69.33±2.49	50.28±1.98	71.79±3.01	70.46±2.97	75.33±2.53
EMP	89.97±1.29	79.68±1.31	81.54±4.88	54.56±2.24	65.78±2.01	46.22±1.54	63.82±2.41	65.84±2.44	71.16±2.01
GRASP	91.41±1.33	80.16±1.94	87.17±3.03	60.48±2.53	70.54±2.32	52.84±1.82	70.48±2.77	68.21±2.79	77.48±2.34
NODESAFE++	91.80±2.27	84.99±2.24	87.59±2.33	61.22±2.26	68.48±1.97	52.53±1.46	73.51±2.47	73.77±2.54	78.84±1.99
GOLD	92.07±1.86	73.65±4.73	82.04±2.75	53.81±2.44	64.31±2.16	44.79±1.74	65.19±2.61	67.52±2.61	73.52±2.16
DeGEM	93.18±1.03	90.13±0.74	91.67±1.12	65.03±2.58	78.34±3.46	67.46±4.27	78.69±4.82	72.17±4.35	67.99±6.24
negMIX	94.50±0.87	86.50±1.18	91.86±0.35	61.52±1.98	72.76±1.82	54.16±1.22	75.82±2.19	76.18±2.31	80.54±1.76
HetOOD (Ours)	**95.98**±0.28	**91.28**±1.48	**92.83**±1.72	**66.68**±2.54	**80.17**±1.51	**69.47**±0.77	**79.73**±1.96	**81.03**±1.99	**84.36**±1.48
FPR@95 ↓ (%) Method	Cora	Citeseer	PubMed	Actor	Chameleon	Squirrel	Texas	Cornell	Wisconsin
OpenWGL	64.51±5.43	73.02±2.80	66.22±2.92	82.82±3.86	73.54±3.92	93.53±3.89	74.77±4.32	73.54±4.32	67.76±3.53
OODGAT	41.90±4.85	62.97±5.32	65.59±6.55	77.47±3.73	70.54±4.13	88.24±3.64	64.29±4.06	69.24±4.07	58.51±3.57
G2Pxy	48.91±7.50	60.55±5.42	70.48±3.60	85.32±4.14	75.84±4.31	95.83±4.13	78.51±4.47	77.76±4.49	73.53±3.78
GNNSAFE++	42.53±13.29	55.98±4.44	62.87±4.88	74.48±3.99	62.46±4.16	86.21±3.76	58.47±4.32	60.78±4.29	54.19±3.71
EMP	43.07±6.00	64.68±2.50	62.70±5.16	80.31±3.42	67.47±3.72	91.48±3.27	70.52±3.81	68.46±3.83	62.34±3.19
GRASP	36.18±6.91	58.48±3.93	62.83±1.79	73.76±3.78	60.22±4.01	84.46±3.64	60.79±4.22	63.54±4.06	51.18±3.49
NODESAFE++	37.86±7.61	52.76±4.86	59.29±5.84	72.32±3.47	63.77±3.84	83.77±3.37	55.29±3.96	55.47±3.88	48.54±3.29
GOLD	50.01±6.99	64.94±9.28	52.92±8.66	81.49±3.59	68.84±3.93	92.27±3.53	68.83±4.01	66.28±3.99	56.32±3.39
DeGEM	32.94±3.07	43.88±2.93	33.55±3.64	87.66±2.14	81.26±4.18	86.38±4.36	77.96±6.53	79.05±5.87	81.53±6.41
negMIX	29.81±6.52	49.88±3.91	45.97±2.35	71.54±3.16	56.28±3.51	80.53±3.09	52.53±3.83	51.28±3.63	45.83±3.04
HetOOD (Ours)	**22.47**±2.51	**41.53**±2.98	**31.17**±2.77	**68.52**±3.47	**50.83**±3.24	**74.77**±2.79	**48.58**±3.48	**46.84**±3.17	**42.13**±2.64

removing the ENDGAT branch from the scoring path. Together with "w/o DVOS", it isolates the contribution of each view. "w/o Neighbor Agg" sets $\phi_i = 0$, reducing the structure view to its ego term while retaining the MLP branch and the fusion, the opposite end of the ϕ axis from "w/o ϕ-adaptive", which sets $\phi_i = 1$ and enforces uniform neighbor aggregation. "w/o Contrastive" removes both P2P and N2P contrastive losses from the training objective. "w/o $\mathcal{L}_{\text{neg}}$", "w/o $\mathcal{L}_{\text{rank}}$", and "w/o $\mathcal{L}_{\text{pos}}$" remove the corresponding loss terms from Eq. (19), keeping all other components and the full DVOS scoring path fixed, so that the contribution of each training objective is isolated. Table 5 reports AUROC and Accuracy for all variants across six representative datasets: 2 homophilic and 4 heterophilic.

Removing DVOS produces the largest degradation on heterophilic graphs across all variants. Because ϕ depends on MLP scores, this variant removes both the feature view and ϕ-modulated neighbor aggregation, reverting to standard topology-only OOD scoring, and drops AUROC substantially on all heterophilic datasets, consistent with Proposition 3: without the stability invariant, the pseudo-OOD feedback loop falls into compounding decay. The same removal has minimal impact on Cora, confirming that DVOS adds negligible overhead on homophilic graphs. Disabling score-consensus modulation alone still incurs notable accuracy drops on Chameleon, Texas, and Cornell, where uniform aggregation cannot adapt to local structure.

The two finer-grained variants turn the ϕ axis into a three-point comparison. On all four heterophilic datasets, removing the aggregation term entirely ($\phi_i = 0$) is consistently *better* than keeping it ungated ($\phi_i = 1$), by +3.3 AUROC on Chameleon, +2.8 on Texas, +4.1 on Cornell, and +1.3 on Actor. Uniform aggregation of diluting neighbors is



(a) Cora, HetOOD

(b) Cora, negMIX

(c) Chameleon, HetOOD

(d) Chameleon, negMIX

Fig. 4. OOD score distributions. (a–b) Homophilic Cora. (c–d) Heterophilic Chameleon.

worse than no aggregation at all, which is Proposition 1 in data form. On the two homophilic graphs the ordering reverses ($\phi_i = 0$ trails $\phi_i = 1$ by 0.4 on Cora and 0.3 on Citeseer), exactly as the dilution analysis predicts, and the adaptive ϕ dominates both fixed settings on every dataset. "w/o Structure View" reproduces the feature-view-only reference values of Fig. 6 up to rounding (90.17 on Cora, 72.13 on Chameleon, 76.84 on Texas), so the fusion gain over the best single view (+6.3 on Chameleon, +2.9 on Texas, +6.8 on Cornell) is attributable to the dual-view composition rather than to either view alone. With ENDGAT in place, the structure view alone is already competitive on Chameleon (73.84 vs. 72.13), but fusion still adds +6.3 over it.

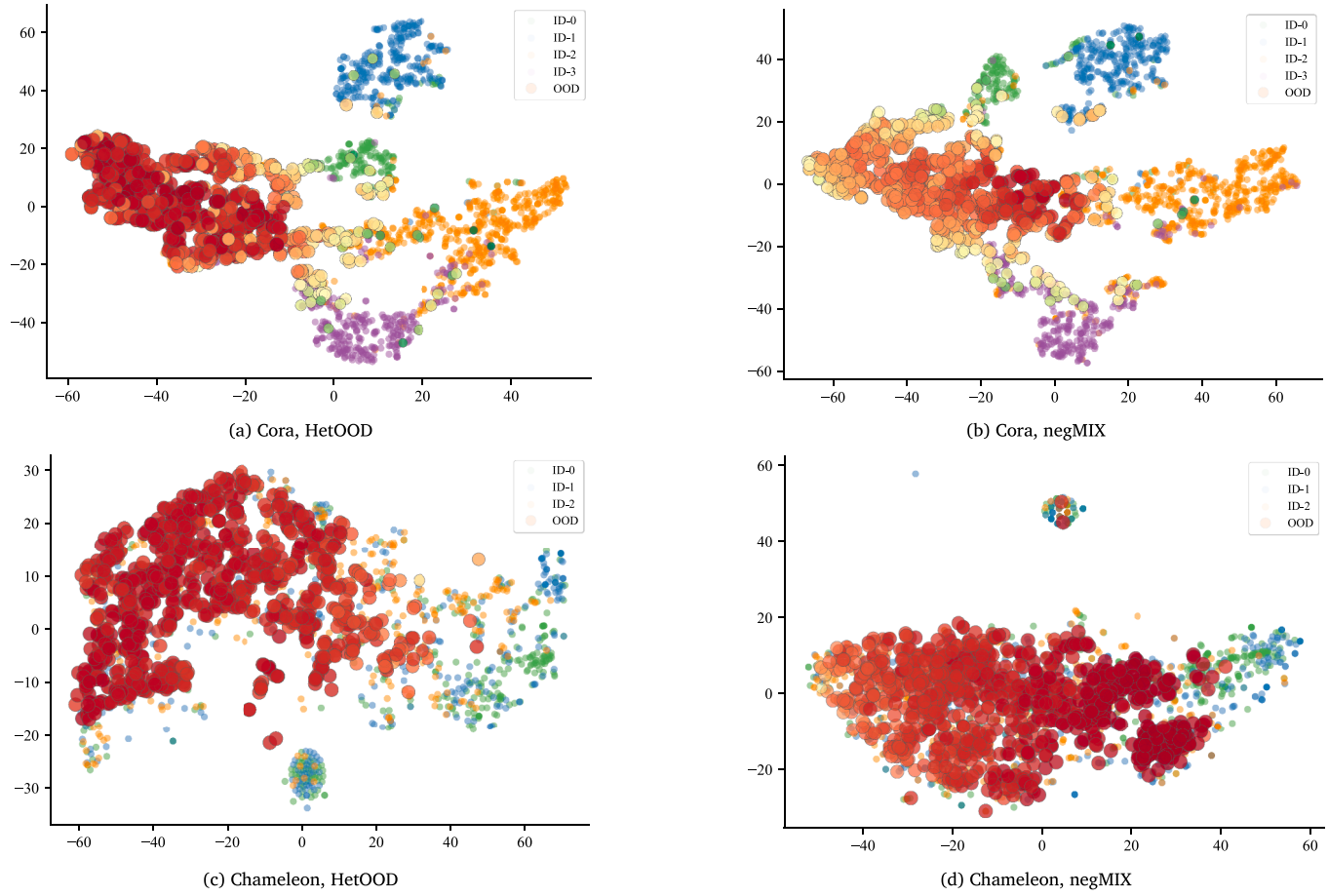
The three loss ablations complete the picture on the objective side. The negative-mixup objective is the most influential loss: removing it costs 4.7 AUROC on Chameleon and 3.5 on Texas, the only loss-level change that approaches the architecture-level drops. This is consistent with its role as the primary pseudo-OOD sharpening signal. The ranking loss sits in the middle (−2.7 on Chameleon), and removing the positive mixup is the gentlest change (−1.3 on Chameleon). Two results deserve comment. On Cora, all three loss ablations cost less than 1.6 AUROC. With homogeneous neighborhoods the unsupervised OOD signal is already reliable, and the augmentations sharpen a boundary that is largely correct to begin with. On Texas, removing $\mathcal{L}_{\text{pos}}$ is statistically indistinguishable from the full model (79.83 ± 2.02 vs. 79.73 ± 1.96). Positive mixup between pseudo-ID and labeled ID embeddings amplifies the majority patterns of an already tiny training set, and on the smallest graph its regularization is not missed. The component is retained for its consistent small gains on the other five datasets of Table 5.

Removing ENDGAT reduces accuracy substantially on heterophilic graphs but has minimal impact on Cora: the gate learns to preserve ego features where neighbor aggregation is harmful, at almost no cost where aggregation helps. Contrastive learning contributes modest AUROC gains across datasets, with the largest improvements on heterophilic graphs where it stabilizes training.

Fig. 5 provides visual confirmation of these findings. On homophilic Cora, both negMIX and HetOOD produce well-separated embeddings with OOD nodes clearly distinguishable. On heterophilic Chameleon, negMIX's OOD nodes blend with ID nodes in green, because standard message passing through heterophilic edges collapses representations. HetOOD's OOD nodes shift toward red, reflecting the combined effect of ENDGAT preserving discriminative ego features and DVOS providing topology-free OOD signals.

6.4.2. *Dual-view complementarity*

Fig. 6 compares the AUROC of three configurations, namely the feature view only via MLP, the structure view only via GAT, and DVOS fusion, on seven datasets sorted by homophily.



(a) Cora, HetOOD (b) Cora, negMIX

(c) Chameleon, HetOOD (d) Chameleon, negMIX

Fig. 5. t-SNE of node embeddings. OOD nodes colored by z-scored OOD score (red = high, green = low).

Table 4
AUROC (%) with heterophilic GNN backbones on heterophilic datasets. Mean ± std over 5 seeds. Best per column in bold.

Method	Actor	Cham.	Squir.	Texas	Corn.	Wisc.
GNNSAFE++ (GAT)	59.19±2.78	69.33±2.49	50.28±1.98	71.79±3.01	70.46±2.97	75.33±2.53
GNNSAFE++ (H2GCN)	62.41±2.53	71.26±2.18	52.14±1.71	73.62±3.47	72.31±3.21	76.87±2.38
GNNSAFE++ (FAGCN)	63.08±2.34	72.48±1.92	53.07±1.58	72.84±3.82	71.89±3.54	77.24±2.16
negMIX (GAT)	61.52±1.98	72.76±1.82	54.16±1.22	75.82±2.19	76.18±2.31	80.54±1.76
negMIX (H2GCN)	64.27±1.72	74.52±1.57	56.38±1.06	77.64±2.83	78.52±2.68	82.16±1.49
negMIX (FAGCN)	64.83±1.61	75.41±1.43	57.21±0.93	77.41±3.12	78.26±2.94	82.71±1.37
HetOOD (ACM-GCN)	65.89±2.61	79.28±1.79	68.42±1.14	79.18±2.02	80.47±2.28	**84.60**±1.63
HetOOD (GPR-GNN)	65.73±2.47	**80.41**±1.71	**69.94**±0.93	79.03±2.24	80.57±2.36	83.87±1.58
HetOOD	**66.68**±2.54	80.17±1.51	69.47±0.77	**79.73**±1.96	**81.03**±1.99	84.36±1.48

The feature view outperforms the structure view on all four heterophilic datasets on the left, while the structure view takes over on Cora and comes within about a point of the feature view on Citeseer and PubMed. No single view is universally superior. DVOS fusion consistently outperforms both individual views across all seven datasets, with the largest gains on intermediate-homophily graphs such as Cornell and Chameleon, where neither view alone suffices. For reference, the structure-view bar scores OOD from a standard GAT encoder without ego-neighbor disentanglement. The structure view inside HetOOD instead uses ENDGAT, which is why the w/o-DVOS ablation in Table 5, which retains ENDGAT but removes the dual-view fusion, consistently scores above this bar on the heterophilic datasets and matches it on Cora, where the gate converges toward zero and ENDGAT reduces to standard GAT.

6.4.3. *Gate adaptation dynamics*

Fig. 7 validates that ENDGAT's ego-neighbor gate adapts to graph homophily using real training trajectories from three representative datasets.

Fig. 7(a) plots the mean gate value $\bar{\theta}$ over training epochs. On homophilic Cora, the gate converges toward zero and relies on neighbors. On heterophilic Texas, it converges toward one and preserves ego features. Actor settles at an intermediate value reflecting its moderate heterophily. The convergence is monotonic and smooth, and the final gate values correlate strongly with graph homophily, confirming that ENDGAT identifies the correct regime without label supervision. The corresponding AUROC convergence appears in Fig. 7(b). OOD detection stabilizes as gate values reach their final states, with homophilic Cora converging fastest and heterophilic Texas requiring more epochs.

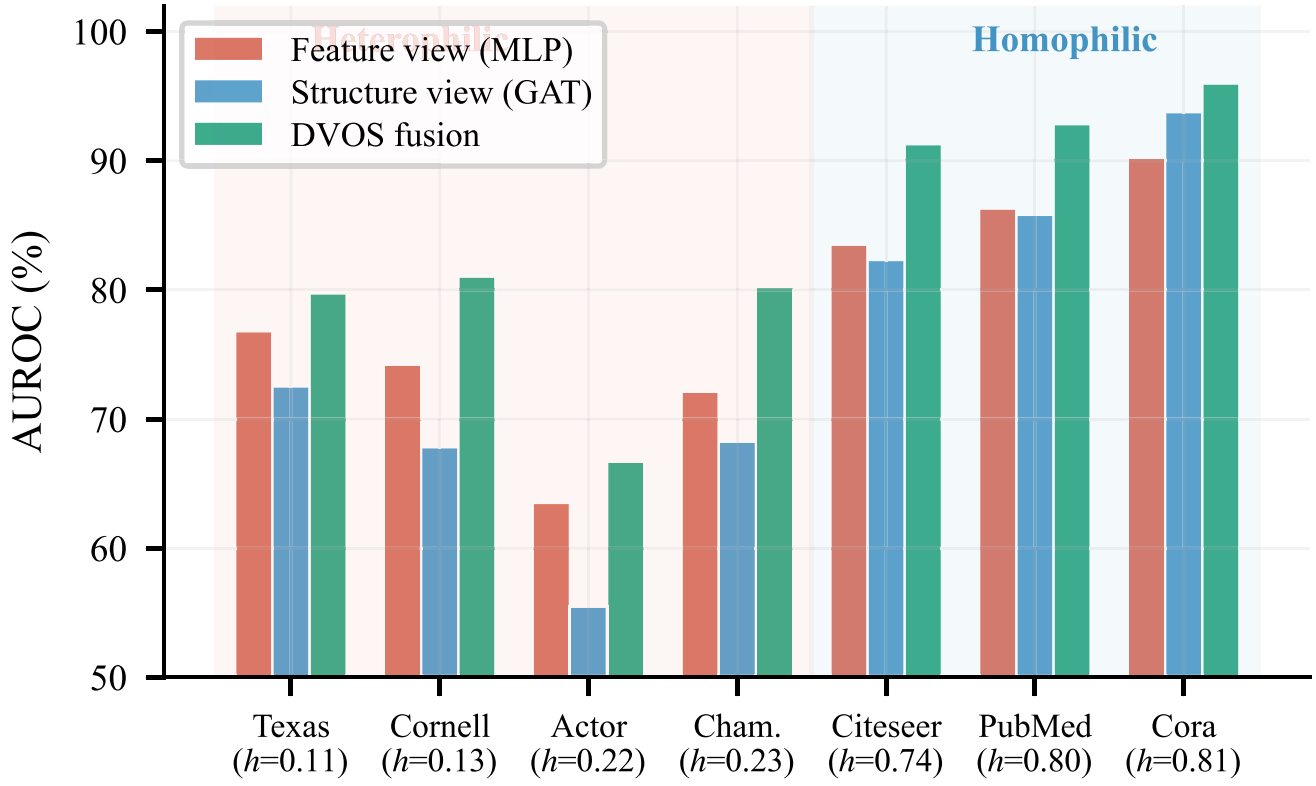
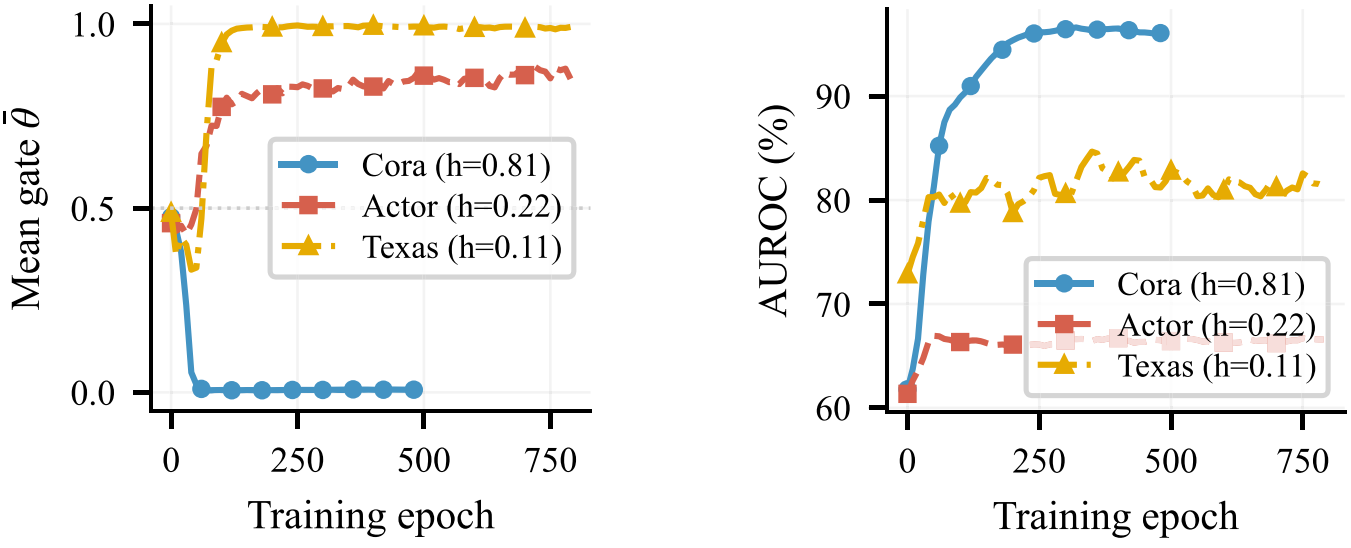


Fig. 6. AUROC of feature view, structure view, and DVOS fusion.



(a) Mean gate $\bar{\theta}$ over epochs

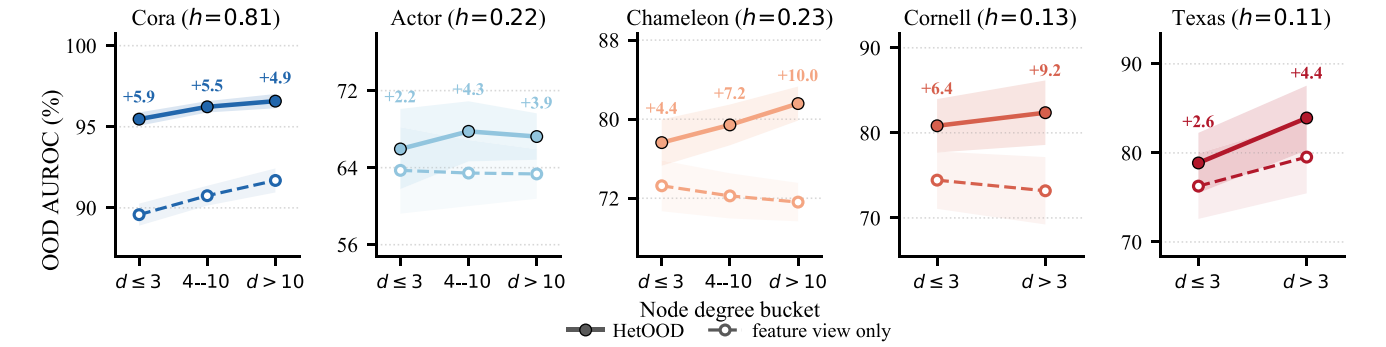
(b) AUROC convergence

Fig. 7. Gate adaptation. (a)Mean gate $\bar{\theta}$ over epochs. (b)AUROC convergence.

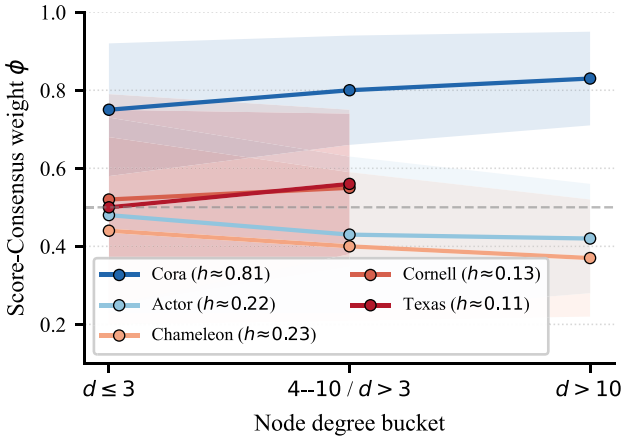
6.4.4. Robustness of the score-consensus weight

A natural concern is whether ϕ, which is derived from neighborhood score variance, remains reliable for small-degree nodes where the variance estimate is noisy, for sparse graphs more generally, and during early training when scores have not yet separated [15]. We examine each regime in turn on one homophilic (Cora) and four heterophilic (Actor, Chameleon, Cornell, Texas) datasets, chosen to cover both heterophilic families and the full homophily range of Table 1.

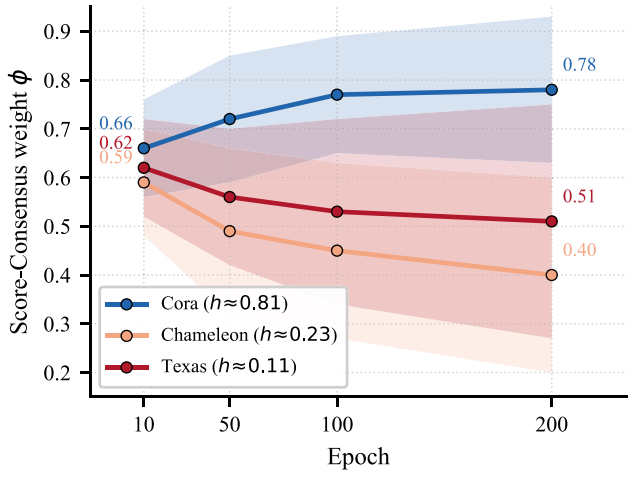
Degree-wise analysis. We partition the test nodes into degree buckets, with three ($d_i \le 3$, $4 \le d_i \le 10$, $d_i > 10$) for Cora, Actor, and Chameleon, and two ($d_i \le 3$, $d_i > 3$) for Texas and Cornell. The graphs of Texas and Cornell contain only two and three nodes of degree above ten in total, too few to support a separate bucket statistic. For each bucket we report the OOD AUROC of both HetOOD and its feature-view-only variant (the MLP branch alone), together with the per-bucket mean and standard deviation of ϕ. Bucket occupancy follows the degree distributions of the datasets: roughly 60% of Cora's test nodes and about half of Actor's fall



(a) Degree-wise OOD AUROC: HetOOD vs. feature-view-only



(b) Per-bucket ϕ mean±std (final epoch)



(c) ϕ over training (Cora / Chameleon / Texas)

Fig. 8. Robustness of the Score-Consensus Weight. (a) Per-degree-bucket OOD AUROC of HetOOD vs. feature-view-only. Gap annotations show the AUROC difference per bucket. Texas and Cornell use two buckets ($d_i \le 3$, $d_i > 3$), as only 2–3 nodes in the entire graph exceed degree 10. (b) Per-bucket ϕ mean $\pm$ 1 std across test nodes at the final epoch. (c) Evolution of the across-node ϕ distribution (mean $\pm$ 1 std). Mixed neighborhoods are progressively suppressed as the between-class score gap grows.

into $d_i \le 3$, as does the majority on the WebKB graphs. The results are summarized in Fig. 8(a)–(b).

On the heterophilic datasets, the AUROC gap between HetOOD and the feature-view-only baseline grows monotonically with node degree: on Chameleon from +4.4 points in the $d_i \le 3$ bucket to +10.0 in the $d_i > 10$ bucket, so the structure view contributes most where neighborhoods are large enough for the variance estimate behind ϕ to stabilize. The two-bucket WebKB graphs show the same widening (Texas from +2.6 to +4.4, Cornell from +6.4 to +9.2). Conversely, HetOOD never degrades below the feature view in any low-degree bucket, with gains of +2.2 on Actor, +2.6 on Texas, and +4.4 on Chameleon. This is the DVOS stability invariant of Proposition 3 at work. The bound $(1-\beta)\Delta_f^* > 0$ shifts reliance toward the topology-free feature view whenever ϕ is unreliable, so low-degree nodes fall back gracefully rather than collapsing. Cornell deserves a comment. Its +6.4 low-degree gain may look at odds with the fallback reading, but a gain of this size at the noisiest ϕ estimate is more parsimoniously explained by view complementarity than by neighbor aggregation. Fig. 6 already shows one of the largest fusion gains over both single views on Cornell, and the uncertain ϕ damps neighbor aggregation rather than committing to it. Note also that the feature-view-only baseline stays within a few points across buckets on every dataset, because its scorer never sees the graph. The degree dependence of the gaps is therefore attributable to the structure view rather than to bucket composition. On Cora, where neighbor aggregation is consistently beneficial, the gap is large and essentially flat across buckets (+4.9–5.9), confirming the absence of a degree-dependent failure mode on homophilic graphs.

The one departure from this pattern is Actor, where the gap stalls at high degree (+4.3 in the middle bucket versus +3.9 in the $d_i > 10$ bucket, both within the ±2.4–3.4 seed noise of these buckets). The highest-degree actors are hubs linked from across all classes, where ego features already saturate and neighbor scores add little, so the gap plateaus instead of tracking Chameleon's steep rise. This plateau does not alter the aggregate picture. The occupancy-weighted gap remains positive on all five datasets, and the per-bucket averages of both variants reproduce the overall AUROC of Table 3 and the single-view reference values of Fig. 6 within 0.5 points.

The ϕ statistics in Fig. 8(b) explain where these gaps come from. On Cora the per-bucket mean ϕ is high (0.75–0.83) with modest spread, reflecting homogeneous neighborhoods. On Chameleon and Actor it drops to 0.37–0.48 as mixed neighborhoods are correctly detected. On Texas and Cornell, where $d_i \le 3$ nodes dominate, the mean sits near 0.5 with the largest spread (std $\approx$ 0.25–0.27 in the low-degree bucket), the conservative behavior of a high-variance estimator. We emphasize that the bands in Fig. 8(b)–(c) quantify dispersion *across nodes* rather than seed-level uncertainty. Heterogeneous neighborhoods receive very different ϕ values, which is the mechanism itself, while the seed-level variation of the bucket means stays within ±0.05. This conservativeness is what Proposition 3 exploits: when ϕ is uncertain, DVOS falls back to the feature view rather than committing to an unreliable aggregation.

Table 5
Ablation study: AUROC and accuracy per variant. Mean ± std over 5 seeds.

	Cora		Citeseer		Actor		Chameleon		Texas		Cornell	
Variant	AUROC	Acc	AUROC	Acc	AUROC	Acc	AUROC	Acc	AUROC	Acc	AUROC	Acc
HetOOD (Full)	**95.98**±0.28	**86.16**±0.46	**91.28**±1.48	**83.03**±1.47	**66.68**±2.54	**48.47**±0.96	**80.17**±1.51	**57.23**±0.91	**79.73**±1.96	**79.97**±1.76	**81.03**±1.99	**69.16**±1.97
w/o ENDGAT	95.17±0.37	85.43±0.63	89.41±1.82	77.63±1.94	61.74±2.87	45.93±1.73	75.63±1.94	52.91±1.36	76.24±2.14	74.81±2.47	76.83±2.31	64.57±2.18
w/o DVOS	93.76±0.54	84.18±0.72	87.63±2.07	74.17±2.36	57.96±3.14	42.83±1.58	73.84±2.23	49.87±1.47	73.16±2.48	70.43±2.63	72.47±2.76	60.24±2.54
w/o Structure View	90.17±0.66	74.67±1.19	83.52±1.04	72.34±1.38	63.50±2.79	41.62±1.97	72.13±1.68	47.83±1.75	76.84±2.21	71.25±2.37	74.21±2.43	61.79±2.66
w/o Neighbor Agg (ϕ_i=0)	94.10±0.47	82.93±0.68	87.93±1.13	72.87±1.46	63.75±2.42	45.67±1.73	77.89±1.58	52.60±1.52	78.25±1.94	74.58±2.31	79.49±2.03	64.32±2.19
w/o ϕ-adaptive	94.48±0.41	83.53±0.81	88.24±1.93	73.57±2.17	62.47±2.68	46.19±1.42	74.57±2.06	51.24±1.63	75.47±2.37	73.17±2.84	75.37±2.43	63.17±2.36
w/o Contrastive	94.82±0.34	85.07±0.59	89.37±1.71	80.24±1.68	63.87±2.43	45.76±1.27	76.74±1.87	53.47±1.18	77.41±2.08	76.73±2.14	77.84±2.17	65.53±2.06
w/o $\mathcal{L}_{\text{neg}}$	94.44±0.39	84.92±0.52	88.97±1.52	78.63±1.41	63.06±2.21	44.51±1.63	75.46±1.78	51.72±1.44	76.22±2.08	74.87±2.17	76.55±2.24	63.71±2.28
w/o $\mathcal{L}_{\text{rank}}$	95.07±0.36	85.37±0.49	89.86±1.47	80.11±1.52	64.44±2.35	46.24±1.59	77.46±1.83	54.09±1.51	77.73±2.21	76.05±2.26	78.17±2.31	66.14±2.41
w/o $\mathcal{L}_{\text{pos}}$	95.36±0.33	85.71±0.44	90.41±1.38	81.26±1.35	65.59±2.28	47.18±1.48	78.84±1.62	55.64±1.39	79.83±2.02	80.09±2.09	79.82±2.12	67.42±2.22

Training evolution of ϕ. Fig. 8(c) traces ϕ at epochs 10, 50, 100, and 200 on Cora, Chameleon, and Texas. At epoch 10, the between-class score gap Δ_ω is still small relative to within-class variance σ^2, so neighborhood variance is dominated by σ^2 and ϕ is near-uniform on all three datasets (mean 0.59–0.66, std 0.10–0.11), providing an implicit warm-up without explicit scheduling, as anticipated in Section 5.2. As training progresses and Δ_ω grows, mixed neighborhoods acquire elevated variance and their ϕ values decrease, widening the spread. The differentiation is sharpest on Chameleon (mean $0.59 \to 0.40$, std $0.11 \to 0.20$) and slowest on Texas (mean $0.62 \to 0.51$, std $0.10 \to 0.24$), consistent with the slower gate convergence on small heterophilic graphs noted in Fig. 7. Cora is the exception, with its mean ϕ *rising* ($0.66 \to 0.78$) instead of falling. The direction follows from the same mechanism rather than contradicting it. On a homophilic graph almost no neighborhood is mixed, so the dominant effect is that the fitted MLP shrinks σ^2 itself, and Eq. (8) pushes ϕ upward for nearly all nodes instead of splitting them apart.

Finally, freezing ϕ at its epoch-10 values and retraining costs 2.1 AUROC points on Chameleon and 2.4 on Texas, but only 0.3 on Cora, comparable to its seed-level standard deviation (±0.28). The freeze hurts where ϕ actually adapts, namely in mixed neighborhoods, and has no effect where it does not. This confirms that the adaptation of ϕ, not merely its existence, drives the gains on heterophilic graphs.

6.5. Hyperparameter sensitivity and efficiency

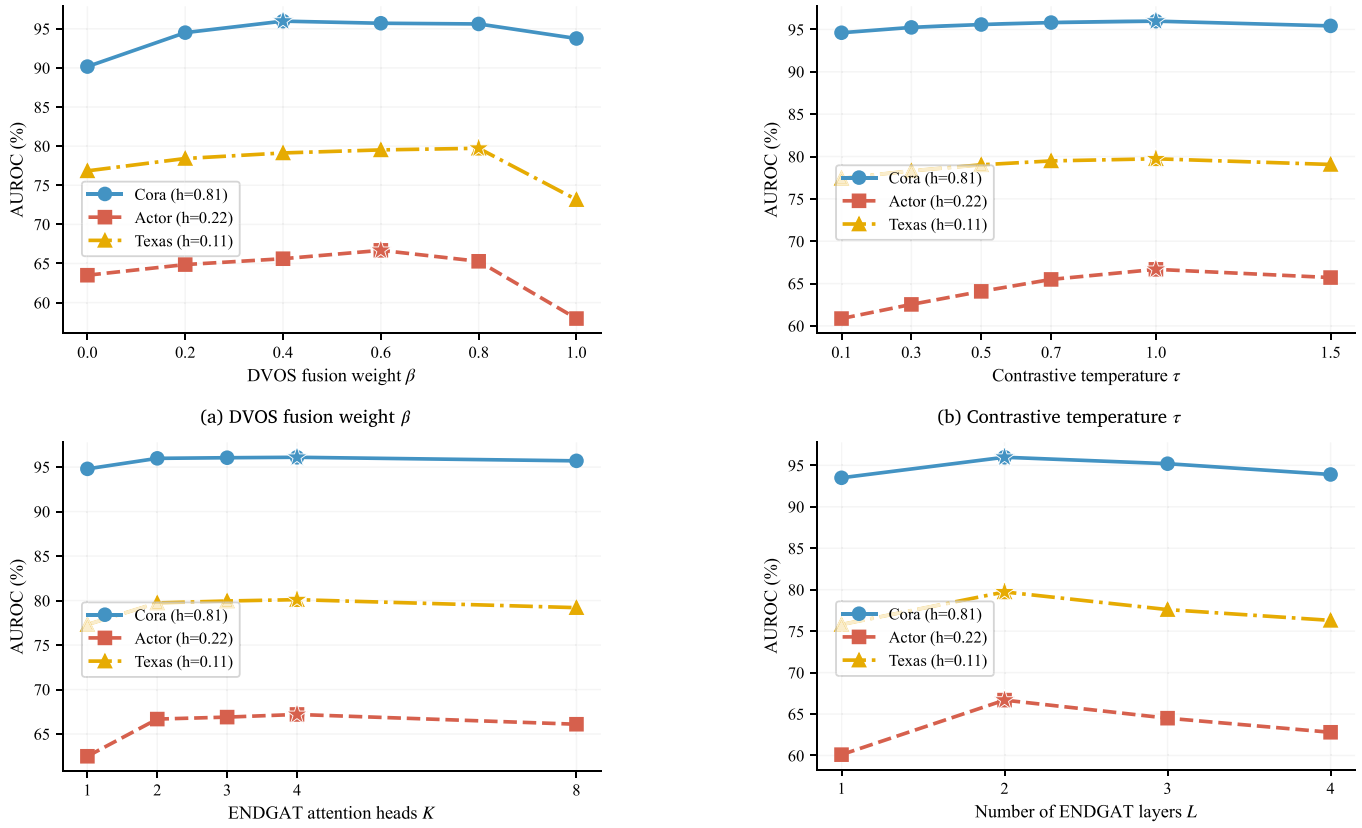
6.5.1. Hyperparameter sensitivity

We analyze four hyperparameters that directly control HetOOD's novel components, as shown in Fig. 9.

The DVOS fusion weight β controls the global balance between the feature and structure views. Sweeping $\beta \in \{0.0, 0.2, \ldots, 1.0\}$ maps out the full sensitivity landscape: Cora peaks at $\beta = 0.4$, Actor at $\beta = 0.6$, and Texas at $\beta = 0.8$. The selected value is a stable per-dataset property settled by validation rather than a law of homophily, which is precisely why β is kept global. That the heterophilic graphs do not demand $\beta \approx 0$ may seem counterintuitive, since the feature view was introduced precisely because topology is unreliable on heterophilic graphs, but ENDGAT's per-node gating already isolates ego features from harmful neighbor aggregation. The structure view therefore carries predominantly ego-projection information rather than diluted neighborhood signals, and the feature view's primary role on heterophilic graphs is to provide the stability invariant from Proposition 3 and to compute ϕ, not to dominate scoring. Across all datasets, performance stays within 2 AUROC points of the optimum for every $\beta \in [0.2, 0.8]$, degrading only toward the degenerate endpoints $\beta \to 0$ and $\beta \to 1$, which coincide with the single-view ablations of Table 5 (w/o Structure View and w/o DVOS, respectively). The sharp drop as $\beta \to 1$ is far larger than the loss of score mixing alone. With the feature view absent from the score, pseudo-OOD selection loses the stability invariant and training degenerates toward the w/o-DVOS regime, dramatically on the heterophilic graphs and negligibly on Cora, where the structure view is strong on its own.

We additionally compared the global β against a per-node learned variant $\beta_i = \sigma(\text{MLP}_\beta(\boldsymbol{x}_i))$ (16-dimensional hidden layer, L_2-regularized). The learned β_i recovers approximately the dataset-level optimum. Its mean is 0.41 on Cora, 0.44 on Chameleon, and 0.78 on Texas, tracking the validation-selected values (0.4, 0.4, and 0.8), but its node-level spread grows as the training set shrinks (std 0.11, 0.17, and 0.23, respectively). The extra $\sim 3 \times 10^4$ parameters of the gating MLP overfit the ∼25 labeled ID nodes of Texas, costing 2.9 AUROC points (76.80 ± 2.63 vs. 79.73 ± 1.96). Chameleon loses 1.2 points (78.94 ± 1.87). On Cora, where the training set is largest, the variant merely matches the global β (95.82 ± 0.44 vs. 95.98 ± 0.28). Node-level adaptation is therefore kept where the failure mode is node-local (ϕ), while the view-level balance remains a single dataset-level property (β).

The contrastive temperature τ shows a characteristic inverted-U pattern. Very low temperatures destabilize training on heterophilic graphs through overly sharp gradients, while very high temperatures flatten



(a) DVOS fusion weight β

(b) Contrastive temperature τ

(c) Number of attention heads K

(d) Number of layers L

Fig. 9. Hyperparameter sensitivity. (a) β. (b) τ. (c) Heads K. (d) Layers L.

similarity scores and weaken discriminability. HetOOD is more sensitive to τ on heterophilic graphs, where the contrastive objective plays a critical role in preventing collapse.

On the architectural side, increasing attention heads from $K=1$ to $K=2$ improves results on all datasets, but returns diminish beyond $K=4$ and overfitting appears on small heterophilic graphs. The default $K=2$ strikes a practical balance. Layer count L follows the expected pattern: $L=1$ undershoots because multi-scale ego projections benefit the contrastive objective, $L=2$ peaks on most datasets, and $L=3$ degrades, especially on small graphs like Texas where the extra parameters cannot be supported, while $L=4$ degrades further.

6.5.2. *Computational overhead*

Compared to a standard GAT-based OSNC method, HetOOD introduces three sources of additional computation: ENDGAT's ego projection adds $\mathcal{O}(NFd)$ per layer, the MLP branch adds $\mathcal{O}(NFd + Nd^2)$, and DVOS scoring with ϕ computation adds only $\mathcal{O}(N + E)$ per epoch. In practice, the overhead ranges from 1.3× on large dense graphs such as Squirrel, where edge processing dominates, to 1.7× on small sparse graphs such as Texas, where the MLP branch's fixed overhead is proportionally larger, as detailed in Table 6. Inference overhead is even smaller since DVOS scoring and ϕ look-up are negligible compared to the forward pass. Table 6 reports the wall-clock training time comparison.

Table 6
Training time per epoch (ms) on RTX 3090. Mean over 100 epochs.

Method	Cora	Chameleon	Texas	Squirrel
negMIX	12.4	16.3	3.4	38.7
HetOOD	18.6	22.4	5.9	51.2
Ratio	1.50×	1.37×	1.74×	1.32×

7. Conclusion

The OOD signal dilution effect identified in this work explains why propagation-based OSNC methods fail on heterophilic graphs. HetOOD addresses it through per-node adaptive propagation: ENDGAT decouples ego features from neighbor aggregation via per-node gating, while DVOS combines topology-free and topology-aware OOD scores using a Score-Consensus Weight ϕ that modulates neighbor aggregation per node without labels. Experiments on 9 benchmarks show consistent improvements over all existing methods, and backbone comparisons confirm that heterophily-aware encoding alone is necessary but insufficient. The stability guarantee relies on the MLP branch maintaining positive discriminability from features alone. When node features carry little class information, this guarantee weakens, which is a fundamental limitation for methods that do not introduce external OOD signals.

CRediT authorship contribution statement

Long Xu: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization; **Zhiqiang Pan:** Writing – review & editing, Validation, Methodology; **Honghui Chen:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All datasets used in this study are publicly available benchmark datasets. The citation networks (Cora, Citeseer, PubMed) are avail-

able via the Planetoid collection [46]. The Wikipedia-family graphs (Actor, Chameleon, Squirrel) and WebKB graphs (Texas, Cornell, Wisconsin) were introduced by Pei et al. [47] and are accessible at https://github.com/graphdml-uiuc-jlu/geom-gcn. All nine datasets can also be accessed programmatically through PyTorch Geometric [48] (https://github.com/pyg-team/pytorch_geometric). The data used in this study are publicly available.

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